Computer-Guided Orthognathic Surgery: State of the Art and Innovation

Alessandro Tel, Tong Xi, and Massimo Robiony

1 Introduction

Orthognathic surgery represents perhaps the most characterizing surgery in the field of maxillofacial surgery. Moreover, it includes the most standardized workflow that has brought orthognathic surgery virtual surgical planning (VSP) also among the less experienced users [1]. This is due to the following reasons:

- Highly sequential surgery with reproducible osteotomies
- Easy, threshold-based segmentation of skeletal bases, and no need to reconstruct soft-tissue volumes except for the facial soft tissue, which can be easily acquired from CT itself of facial scanning
- Need for high precision in reproducing the desired occlusion with planned movements, as each millimeter can have a substantial impact on both functional and esthetic outcomes
- Very easy and scalable surgical planning: osteotomies are linear and can be performed by joining points placed by the user; movements can be chosen from a visual interface implementing simple commands for each spatial direction
- Creation of surgical splints that can be manufactured in-house using commercially available 3D printers
- Entirely incorporated within single software packages available for clinicians and which do not require any prior engineering or CAD knowledge background. This is not the case with other complex simulations including cranial reconstruction, orbital surgery and trauma.

This chapter will focus on the methodology of virtual surgical planning in orthognathic surgery, describing the key sequence ideal to achieve a predictable and reproducible VSP. Emphasis will be placed in reinforcing the connections between virtual actions and their impact on the final outcome of the patient, through a sophisticated cephalometric analysis. The following chapter will demonstrate the application of this methodology in a multitude of clinical scenarios.

2 Prerequisites for Computer-Guided Orthognathic Surgery

Compared with other simulated surgical procedures that traditionally require exclusively to import DICOM data of volumetric imaging, orthognathic surgery needs the following items in order to be performed:

- CT scan: Differently from other surgeries, for orthognathic surgery the patient often undergoes a cone beam CT scan (CBCT). Cone beam CT scan is a volumetric imaging, but it is performed using a low voltage, which is particularly suitable for younger patients. However, compared with traditional multidetector CT (MDCT) usually performed in a hospital setting, CBCT provides a slightly inferior image quality, often presenting a “patchy” texture in the background, as the result of the different features of the x-ray beam implemented. Moreover, traditional Hounsfield Units (HU)-based thresholding intervals for segmentation of bone in MDCT are not applicable to CBCT. Thus, discrimination between bone and nearby tissues is less efficient; moreover, thin or cancellous-bone

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structures, such as orbital walls, condyles and the posterior regions of the maxilla, are often inadequately captured by the thresholding because the intrinsic contrast of voxels is overall inferior. CBCT should include in its acquisition field the orbital region down to the menton (ME) point, extending posteriorly to include both temporomandibular joints. The protocol for CBCT scan should include a 512 × 512 px matrix and a 0.6 mm slice thickness. The CT has to be performed while keeping the condyle in centric relationship, preferably with closed mouth and maximized intercuspation or eventually by interposing a thin wax layer to create a minimal intercuspal disclusion, which does not affect significantly centric relationship and facilitates the separation of dental arches in VSP.

- Dental models: The second essential prerequisite to perform an orthognathic surgical planning is the availability of models of dental arches. Depending on the orthodontist, the patient might arrive to the surgeon with physical plaster models or an intraoral scan.
  - Physical plaster models are yielded by pouring plaster within an alginate impression taken from the patient. They have the advantage to convey the sensation of occlusal stability by placing dental cusps into optimal engagement (intercuspation). The drawback is that such models need to be digitized, can be lost or subject to rupture. Their overall quality is inferior to acquisition using a dental scanner (Fig. 1). Moreover, to replicate the preoperative occlusion, the acquisition of an occlusion wax is necessary.
  - Intraoral scanning is performed using a scanner whose tip can be easily inserted within the oral cavity, and it contains a system of deflection mirrors to capture the shape of each dental cusp from different angulations. Intraoral scanners are based on pulses of a structured light beam, which is reflected and captured by a sensor. The time of reflection of light across each point of the surface is converted into a geometrical output, retessellated and rendered as an STL file compatible with design software. This method can prescind from an occlusion wax because the software is able to acquire the digital landmarks for habitual occlusion and transfer them to the dental arches STLs (Fig. 2). Before the advent of automatic surgical occlusion algorithms, it was hazardous to search for a stable occlusion using only digital models.
  - A mixed solution consists in taking the digital teeth impressions using an intraoral scanner, 3D printing the models and put them manually in the desired surgical occlusion (Fig. 3). This method combines the versatility and safety of an entirely digital workflow with the tactile sensation and the perception of occlusal stability that the intercuspation of plaster models provides. Once the physical occlusion is determined, the intraoral scanner can be used to acquire a registration mask on the plaster models, which is then imported in the virtual project and used to align the intraoral scans of the superior and inferior arched previously acquired. This third solution is useless if virtual occlusion is used.

Additional data can be imported in the virtual project to improve the final virtual plan:

- Soft tissue model: A full volumetric reproduction of soft tissues, yet static, is essential for certain phases of VSP, including natural head positioning, midline evaluation and postsimulation morphing. It is generally achieved through segmentation of CT scan, but it can be substantially improved using a facial scanner. Using a structured light beam, it collects a point cloud, which can then be tessellated into a geometrical triangulated file stored as an STL or OBJ files, or in the more advanced PLY format, which can store information about colors, surface normals and texture coordinates.

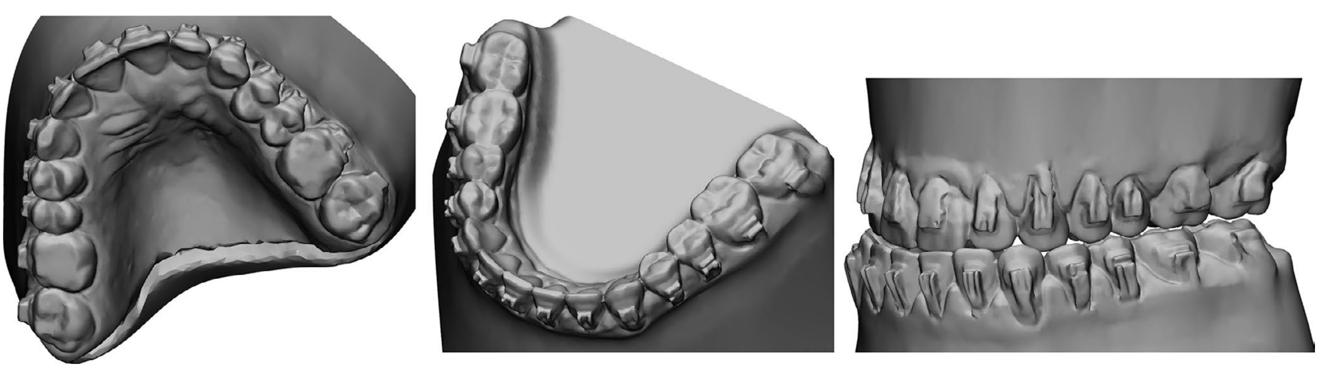


Fig. 1 Scanned plaster models taken through alginate impressions are used to determine the most stable intercuspation. Occlusion is scanned and imported in the VSP software

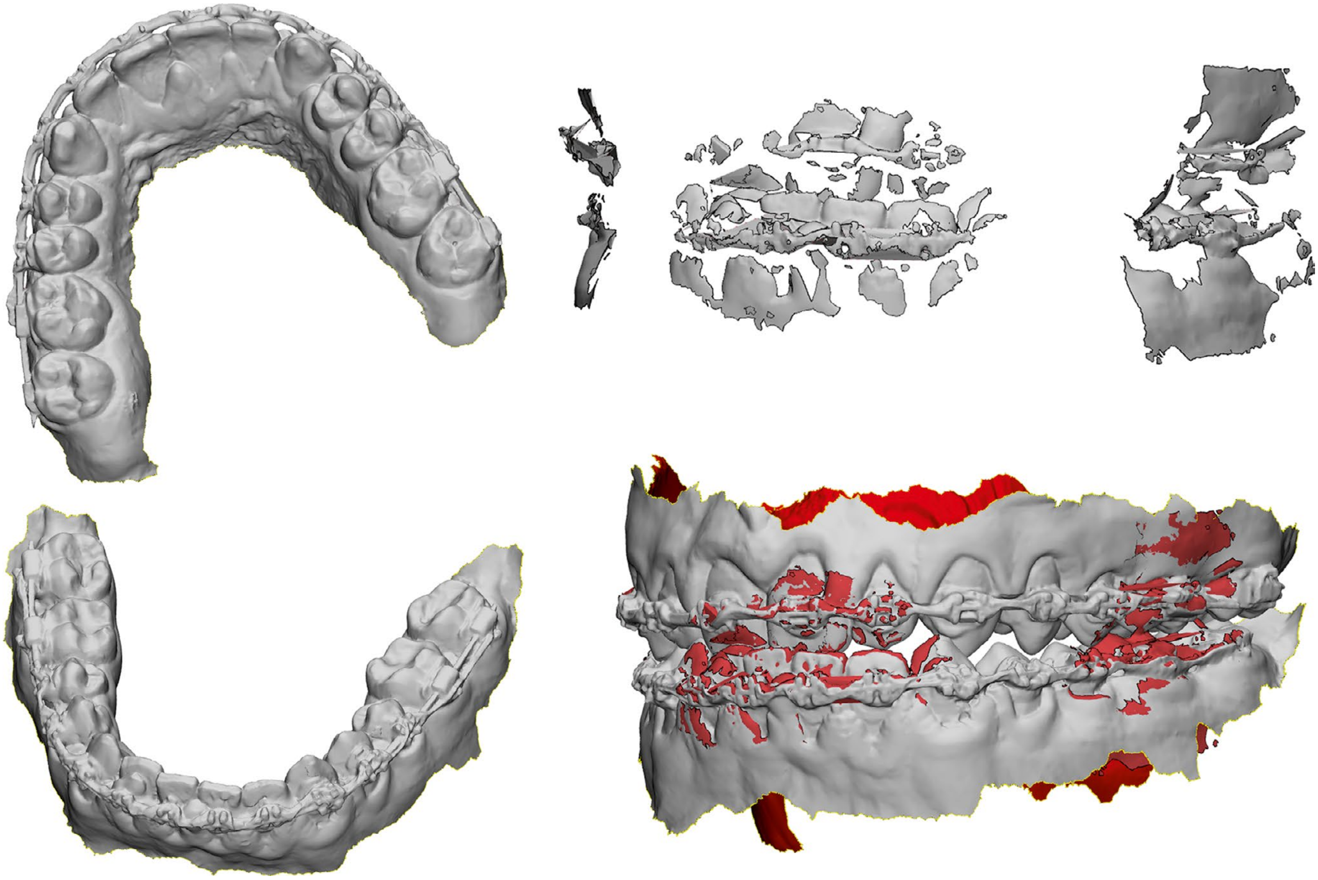


Fig. 2 Intraoral scan shows a consistently higher detail. A registration mask can define the reciprocal alignment of scanned arches

Fig. 3 3D printing of scanned impressions is used to manually determine the most stable occlusion, which is then rescanned and imported in the VSP

Similar sources can be represented by 3D photography or stereophotogrammetry. However, in the majority of cases, a satisfactory CT scan, both MDCT or CBCT, can provide a trustful representation of soft tissue anatomy.
- Airway model: The posterior air space is reconstructed in case a functional evaluation is needed. This is most often performed in cases of severe micrognathia or obstructive sleep apnea.

3 Sequences of Phases for Orthognathic Surgical Planning (Simple Bimaxillary Surgery)

Although the sequence of phases for virtual surgical planning can slightly vary between different software, considering the tendency of newer software to be more "clinician friendly", it is possible to identify a uniform workflow. Due to its widespread presence in hospitals and mentions in scientific literature, in this chapter the Author will describe the basic workflow in the software ProPlan CMF v.3, and its successor, Mimics Enlight (Materialise NV, Leuven, BE).

3.1 Segmentation, Mask Cleanup and Refinement

As previously explained, CBCT are suboptimal compared to MDCT, and a basic segmentation using a single thresholding algorithm would yield models unsuitable for virtual surgical planning (Fig. 4). Starting from the concept that average threshold values for Hounsfield Units are not applicable to CBCTs, and that the intrinsic contrast is lower due to an increased partial volume effect in CBCTs, there are several problems generally encountered [2]:

- CBCT underestimates regions with a predominance of cancellous or rarefied bone, typically in the mandibular condyle and in the posterior sectors of the maxilla, in the tuber region.
- CBCT underestimates regions with thin bone, in particular the nasal cavity, the orbital walls and the palatine bones and pterygoid plate regions due to an increased partial volume effect. This underestimation is much more evident than in MDCT where likewise orbital walls often present interrupted.
- Radiological density of bone and certain regions of soft tissues might overlap: increasing threshold to reduce soft tissue interposition leads to a conspicuous loss of bone, while decreasing it to acquire thinner bone involves conspicuous overlap of non-skeletal voxels.
- Metal artifact is always present in orthognathic patients due to metal appliances used for orthodontic treatment.

Manual mask editing remains the mainstay of image processing to remove unwanted parts. Unfortunately, very few semiautomated tools can assist in this process, which is entirely user-dependent. A solution for loss of bone might be to apply different local thresholds, enabling diverse regions to be captured with different values. This is highly useful in the condylar region, given the importance in orthognathic surgery to visualize the condylar morphology and position. In this case, special segmentation brushes can assist the filling of voids in less radiopaque regions (Fig. 5).

Metal artifact is an always present issue in orthognathic surgical planning, as patients perform the preoperative CT for surgical planning at the end of orthodontic treatment with orthodontic appliances, whose metal component is responsible for X-ray scattering, evident on images as hyperdense beams producing disruptive artifacts on segmentation. Multiple solutions are available to solve this issue, including:

- use of a dual-energy CT machine with modulation of the X-ray energy over metal surfaces [3, 4]
- application of a metal artifact compensation filter, an image-editing feature available in Mimics software, which enables to process the DICOM dataset by applying a given algorithm to reduce the effect of artifact on surrounding anatomy

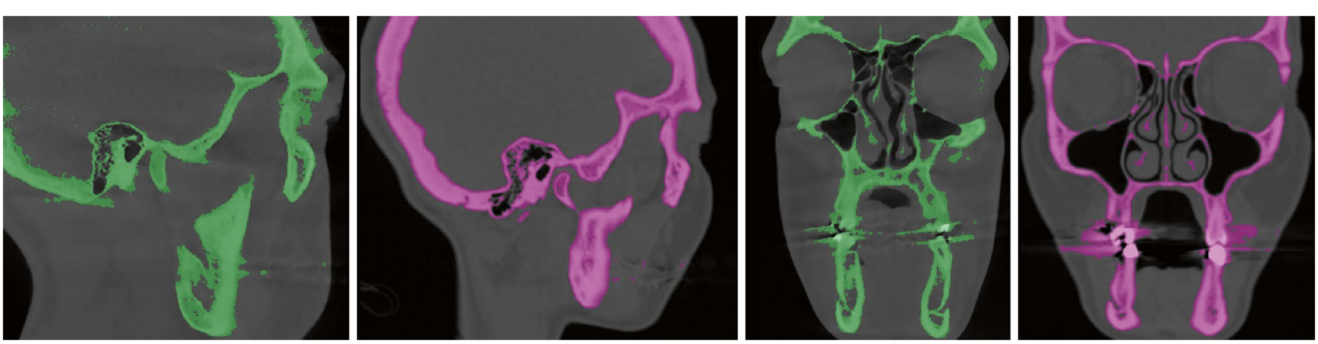


Fig. 4 Differences in bone segmentation by thresholding from a CBCT (green, suboptimal) and a MDCT (purple, optimal)

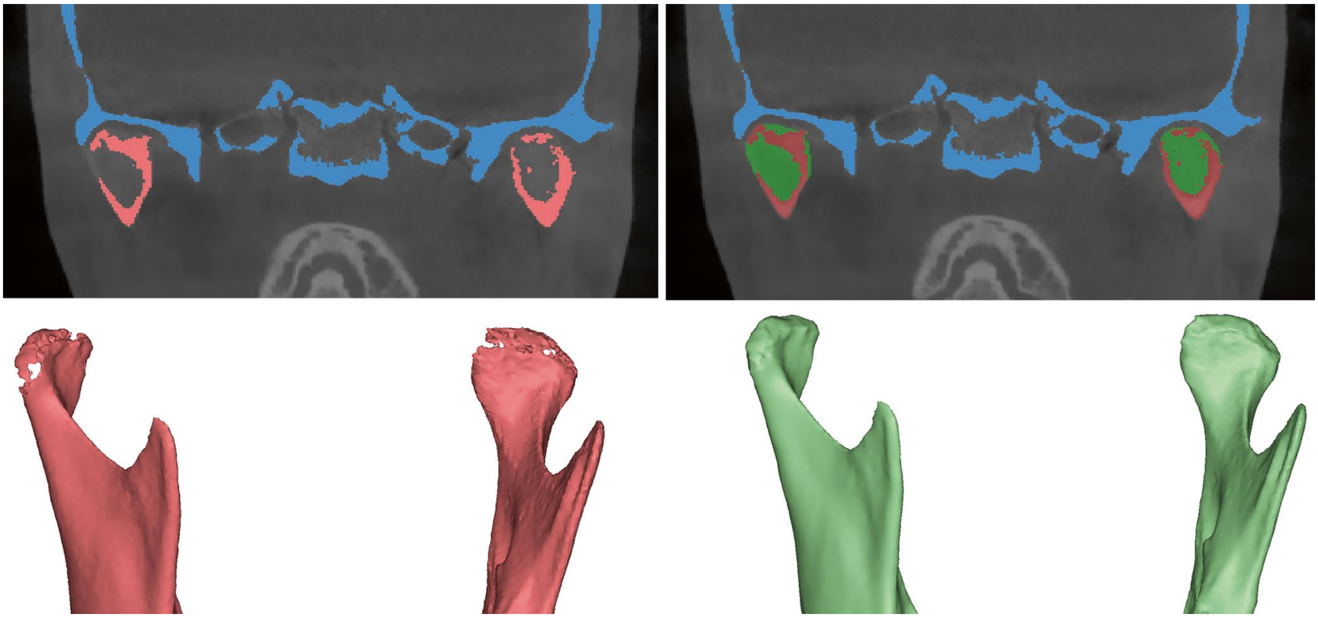


Fig. 5 Smart filling technique used to reconstruct voids in condylar morphology due to the predominantly cancellous structure of the mandibular condyle

- mask editing in correspondence of scattered beams, trying to eliminate them from the 3D models and preserving as much as possible the dental anatomy
- sculpting of the reconstructed polymesh with cleaning of geometry correspondent to scattering areas

It can be noted that the four methods intervene in sequential steps of the process (image acquisition, image editing, mask editing, polymesh editing).

In addition to skeletal model creation, it is advisable to create a segmentation mask for soft tissues. This is generally well achieved by setting a comprehensive threshold that captures the soft tissues and everything inside, except for the airways.

3.2 Separation of the Mandible from the Skull Base

The reconstruction of the 3D model yields a unique solid object, whether it is a segmentation mask or a polymesh, representing the skeletal anatomy of the patient. However, for orthognathic surgery to be simulated, it is essential to provide two distinct models for the skull/maxilla and for the mandible. As the software is unable to distinguish between each of them, it is essential that the user splits the geometrical entities across their reciprocal fusion points: temporomandibular joint and occlusal contacts.

The author suggests that the separation process is accomplished using segmentation masks and not polymeshes, yet it is theoretically feasible for both. Segmentation masks provide a much more flexible environment, where editing operations are easier and not linked to vertex separation. In particular, the splint mask tool designed in Mimics provides the fastest and most accurate solution, as the user defines few splitting interfaces in regions of anatomical contact (in this case, TMJ and dental cusps) and the software is able to interpolate very accurately the input over all slices of the DICOM stack, by recognition of anatomically distinct regions across radiotransparent lines (cusp-to-cusp contact, joint, suture between two bones). The result is the original mask being split in two, one for the mandible and one for the skull/maxilla, and the two masks can be tessellated into different polymeshes (Fig. 6).

3.3 Alignment and Merging of Dental Casts over Skeletal 3D Models

This step is essential because CT scan is not accurate enough to trustfully reproduce the detailed anatomy of dental cusps, in which even submillimeter particulars might have a clinically relevant outcome on the final occlusion. The solution for this problem is to use scanned dental anatomy and register it over the CT-derived 3D model. Scanned dental anatomy, as previously explained, can originate either from optical scanning of a plaster model taken from alginate impressions or can be the result of intraoral scanning, this latter method being the most accurate. Digital impressions are imported into the virtual planning software as STL files, and cusps are aligned with the teeth visible in the CT model using a point-to-point registration, which is then refined

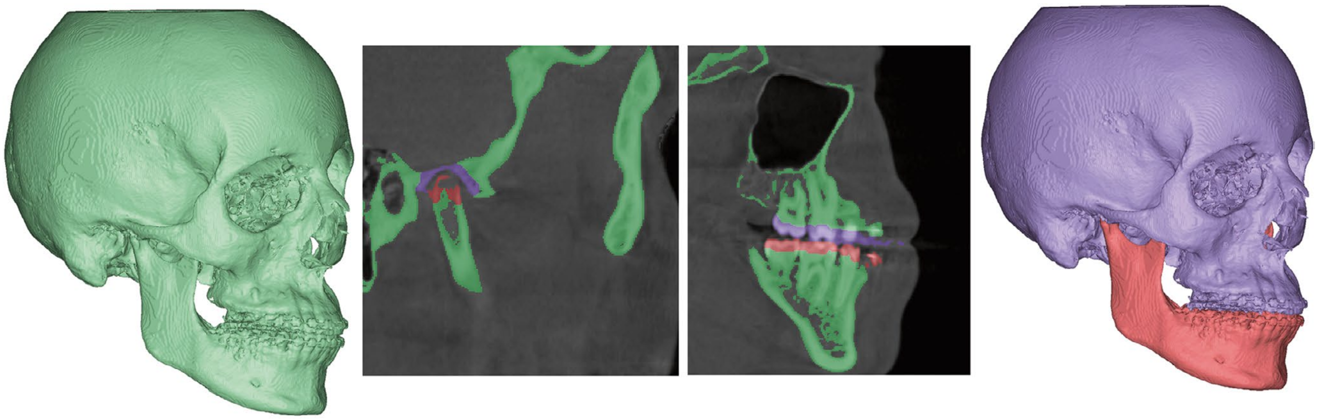


Fig. 6 Mandible separation using the mask split technique applied to contact points: the TMJ and occlusal surfaces

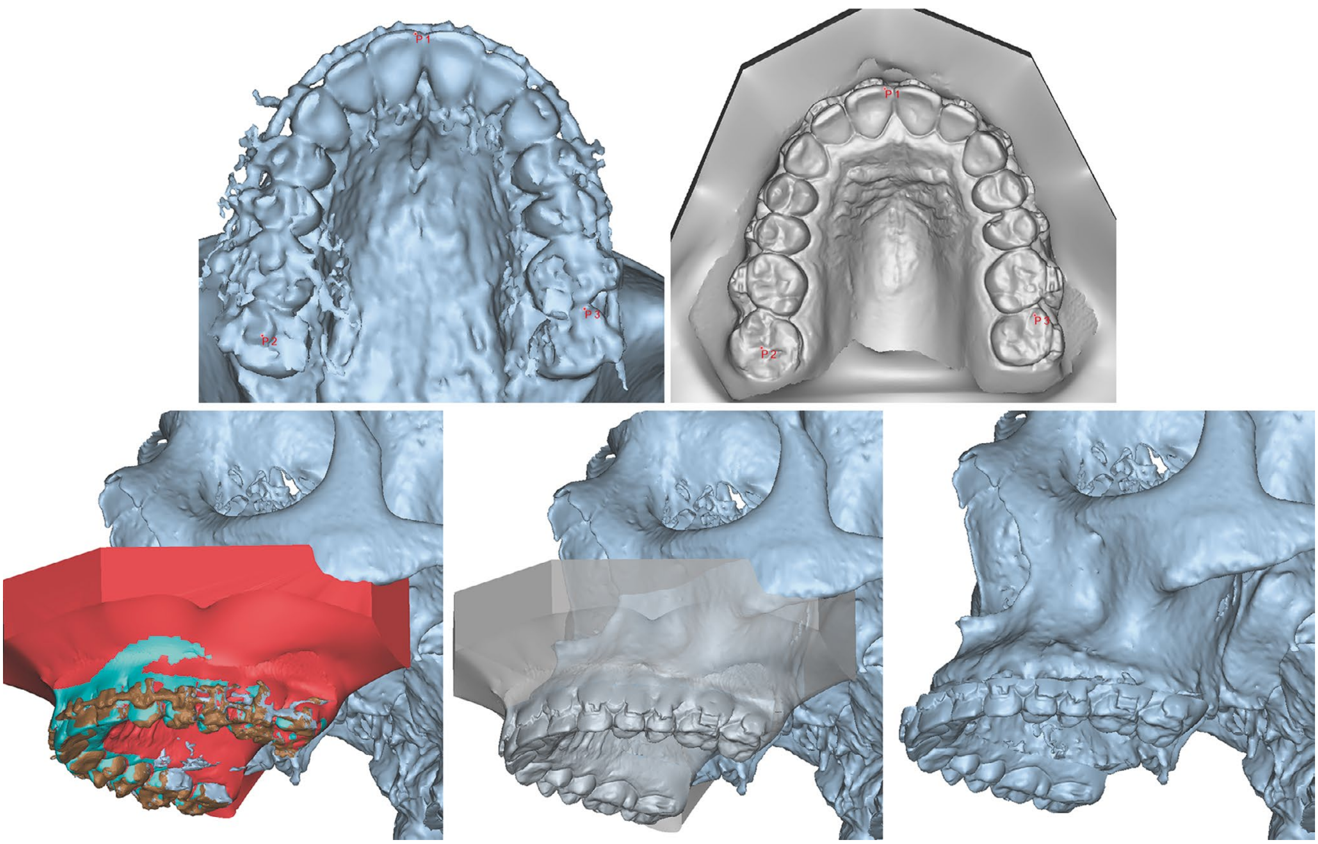


Fig. 7 Registration and fusion between skeletal model reconstructed from CT and scanned impressions (in this example, from plaster models). The less accurate the dental arch in the native CT model, the less performing the alignment process

using a global registration function based on the iterative closest point algorithm.

It is worth mentioning that the registration is performed between two geometrically very different objects, because dental arches reconstructed from CT are much rougher, often not all cusps are visible, and are impaired by the effect of metal artifact. Thus, the quality of the CBCT image is of prominent importance to find the exact matching points between CT cusps and their correspondent cusps on the scanned STL. Once the registration is complete, the software elaborates the merging of geometries, yielding a compound model, which reproduces skeletal surfaces but bears the teeth detail of scanned cusps (Fig. 7).

3.4 Head Orientation

Determining the correct natural head position (NHP) is still an unsolved issue [5–7]. NHP is highly variable between individuals due to a multitude of factors, including postural, proprioceptive, anatomical and psychological. Several methods are documented in the literature to replicate reliably the NHP within the VSP, including the acquisition of CT scan with fiducial markers and a laser to record the horizontal plane or the use of a digital gyroscope attached to a bite fork and a face bow, which provides the quantification of Eulerian angles (pitch, roll, yaw), which can then be transferred to the models.

In the absence of such technologies, NHP can be assessed through accurate photographic documentation. The most effective method to transfer it in the VSP includes the definition of a grid with a vertical axis and a horizontal axis. The horizontal axis should be maintained parallel to a less variable anatomical landmark. For this purpose, the author of this chapter adopts the supraorbital line, rather than the inferior orbital line, which can be more subject to variation especially in cases of skeletal asymmetry. However, it should be noted that severe asymmetries, especially craniofacial malformations requiring orthognathic surgery, might present considerable orbital dystopia affecting the supraorbital line as well. In such cases, more posteriorly placed landmarks should be considered, for instance, a line joining both mastoids. The alignment over the horizontal line accounts for the correction of the roll.

Pitch rotation can be defined by careful assessment of side photographs. Some postural behaviors might conceal the true NHP; for instance, it is common for III class patients with marked mandibular prognathism to elevate the chin in side view. The use of Frankfurt plane has the benefit of standardizing pitch angle but does not correspond to the truth. Therefore, it is advisable to manually remodulate head pitch with careful soft tissue model rotation searching the highest correspondence with clinical photographs.

Concerning yaw, this is probably the most difficult angle. Once roll is corrected, the vertical midline should be made coincident with the true soft tissue midline to evaluate the deviation of the superior interincisal line. This is achieved by correcting the yaw from the bottom view, ideally placing the vertical line as symmetry plane. This should in turn entail that the grid midline, shown in pink in the following illustrations, coincides with the facial midline. In cases in which a mismatch between the position of the soft tissue nasion and the midpoint of the superior labial philtrum, such as in cases of asymmetry, the latter should be preferred owing to its major impact on the patient's smile. In other circumstances, where a considerable lateral repositioning of the maxillo-mandibular complex is planned, the true facial midline might be considered the average between the soft tissue nasion and the midpoint of the superior labial philtrum.

Figure 8 shows the process of reorienting both the skeletal models and the soft tissue according to the NHP.

3.5 Creation of 3D Cephalometry

Once skeletal models with the detail of scanned dental anatomy are available as 3D polymeshes, it is important to determine three-dimensional cephalometric points at their $x_0y_0z_0$ values, which will be modified based upon the planned movements [8]. Software for orthognathic surgical planning generally facilitates this task (Fig. 9) and may incorporate a variety of cephalometric models that can be applied to the project, but the author of this chapter advises the following to be strictly taken into account in an orthognathic VSP:

- Points for dental exposure evaluation: UIL, UIR, UIM (Upper Incisor left, Right and Midline), UCL, UCR (Upper Canine Left and Right). Such points have a critical impact in determining the smile exposure of crowns and are primarily assessed for the esthetic evaluation.
- Points to determine posterior dental movements: 2UML, 1UML, 2UMR, 1UMR (Second and First Molar Left and Right). Such points enable to understand the effect of rotational movements over teeth, such as roll (cant) correction. They can be useful to assess the esthethics as well in the buccal corridors.
- Points to evaluate the skeletal movements of the anterior maxilla: A, ANS (Anterior Nasal Spine). They are useful to assess the effect of vertical movements (impaction) and pitch correction. Their values can be different from dental movements in case of clockwise/counterclockwise rotation.
- Points to evaluate the skeletal movements of the posterior maxilla: PNS (Posterior Nasal Spine). In case of pitch corrections, they provide values opposite from ANS and A.
- Points to evaluate the effect of sagittal mandibular repositioning: B and Pog (Pogonion). They provide the magnitude of advancement/setback.
- Points to evaluate the effect of mandibular asymmetry correction: GoL and GoR (Gonion Left and Right). Such points are especially important if a simultaneous condylectomy is planned.

Mandibular dental points are not included because they depend on the maxillary repositioning. In the VSP phase of skeletal repositioning, the maxillo-mandibular complex in occlusion is considered a unique entity, and esthetical considerations are based upon upper dental points. However, at

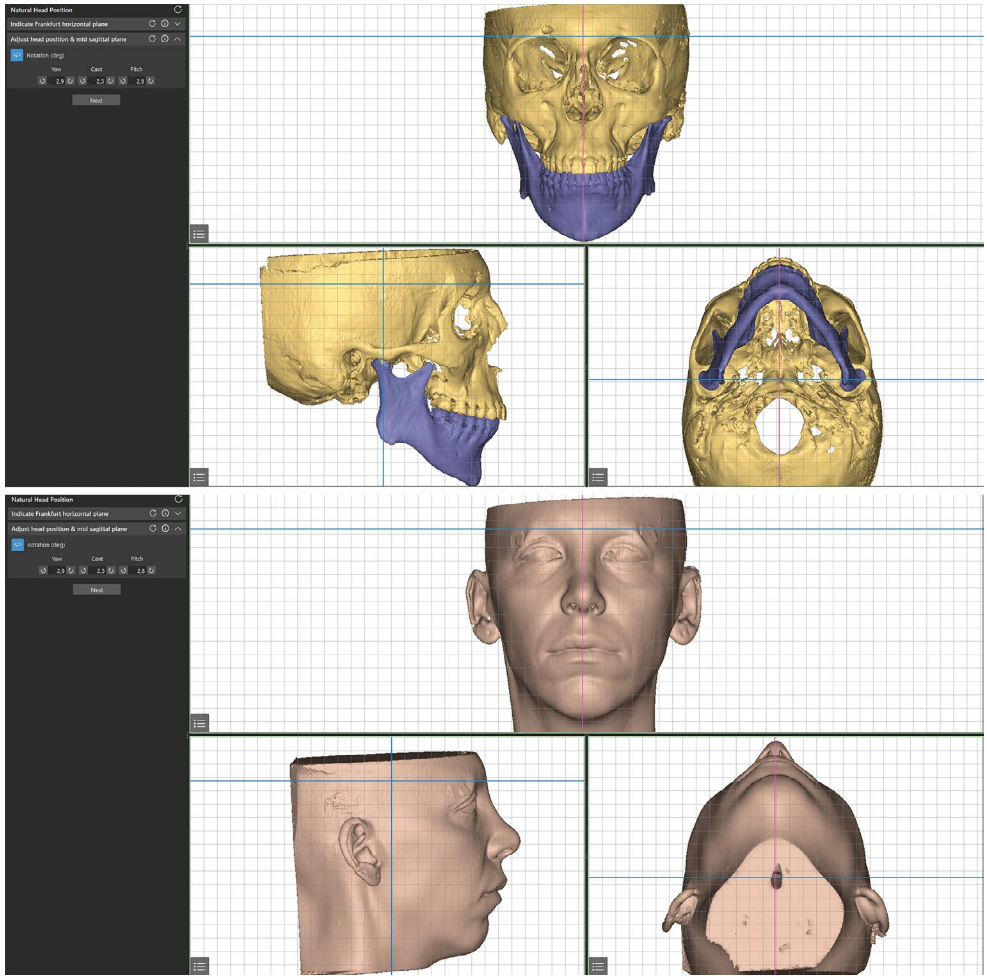


Fig. 8 Natural head position (NHP) adjustment in the software Enlight CMF. Angular degrees are provided. Evaluation should be performed both for the skeletal and the soft tissue model

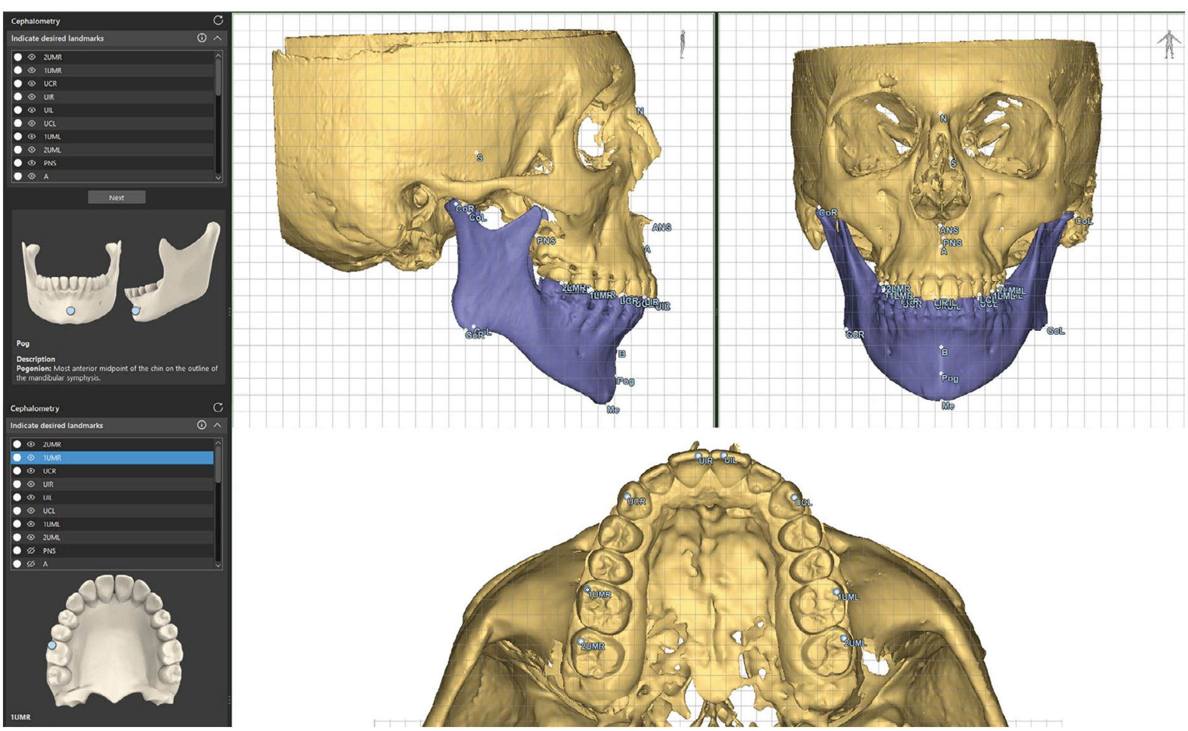


Fig. 9 3D cephalometry guided by visual landmarks in Enlight CMF

the end of VSP, also mandibular dental points values can be inspected to verify the movements of the inferior incisors and canines.

3.6 Osteotomies

Software designed for orthognathic surgical planning, as mentioned, is clinical friendly, and this translates on the fact that designing osteotomies is an easy and fast process, as orthognathic surgery software already provides specific cursors which can be placed over anatomical landmarks to achieved standardized cutting planes.

During segmentation, definition of dental anatomy and teeth roots should be preliminarily accomplished, to establish the osteotomy line far from root apices to avoid their damage. Recent software can make this step automatic, as a volume-rendering algorithm is applied during the creation of the osteotomy plane to emphasize the position of roots, which are typically hyperdense compared to surrounding alveolar bone. The polyplane for osteotomies is designed by defining its origin points and is assigned a thickness, which should generally correspond to the magnitude of the cutting instrument. First, Le Fort I osteotomy plane is defined taking care to cut through all the maxillary bone, especially in the area of the pterygo-maxillary junction. Its osteotomy planes do not pass through the entire geometry, the Boolean operation applied to separate the cut part will result ineffective. Second, before performing the bilateral sagittal split osteotomy of the mandible (BSSO), it is useful to simulate the course of the inferior alveolar nerve, although it can be easily inferred from CT in coronal section (Fig. 10).

After creating osteotomies, the object list will include the following parts: skull, Le Fort I maxilla, both mandibular

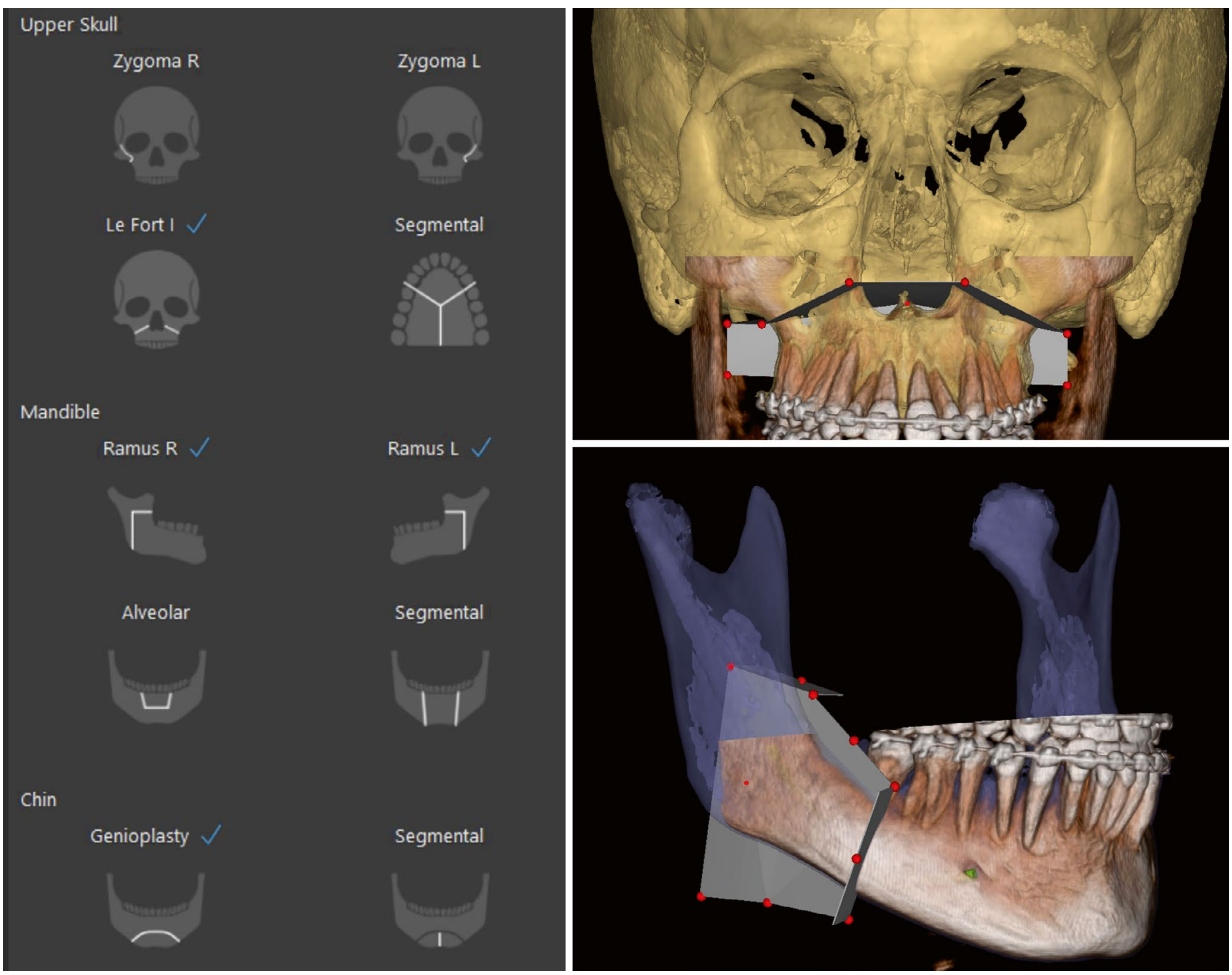


Fig. 10 Le Fort I and BSSO osteotomies with volume-rendered dental roots

rami (proximal mandibular segments), and the distal tooth-bearing segment. Each segment should be assigned a different color and an univocal nomenclature. The same applies to any additional osteotomy performed, including genioplasty, border contouring, multipiece maxilla. Once osteotomies are performed and children's parts are generated, each cephalometric point belonging to the entire geometry will link to the children segments always in the position $x_0y_0z_0$.

3.7 Surgical Occlusion

A crucial step in orthognathic surgical planning is the reproduction of surgical occlusion. There are basically two approaches, yet they are conceptually very different:

- Import occlusion cast: in this method, the occlusion determined by the clinician based on plaster models physical intercuspation (Fig. 11). Occlusion is scanned and is used as template to realign the superior and the inferior dental arch in the correct occlusal relationship within the software. It is a trusted method because it still relies on the traditional approach in which the surgeon determines by his hands the correct occlusal relationship, however its reproducibility in the virtual environment depends on the accuracy of registration, therefore it is not exempt from errors. Moreover, it entails a further scanning of plaster models, which can introduce inaccuracies.
- Automatic virtual occlusion: newer software being developed is increasingly incorporating this function. In this approach, a computational algorithm determines the optimal intercuspation between the superior and the inferior scanned models. The algorithm can apply three contact points, to enhance the automatic intercuspation of dental arches in a tripodic occlusion, or a single dental contact to improve user freedom and decide, for instance, the amount of posterior disclusion. Then, the surgeon can modify the virtual occlusion according to his preferences by acting on specific features, including correction of midline deviation, axial rotation and over-

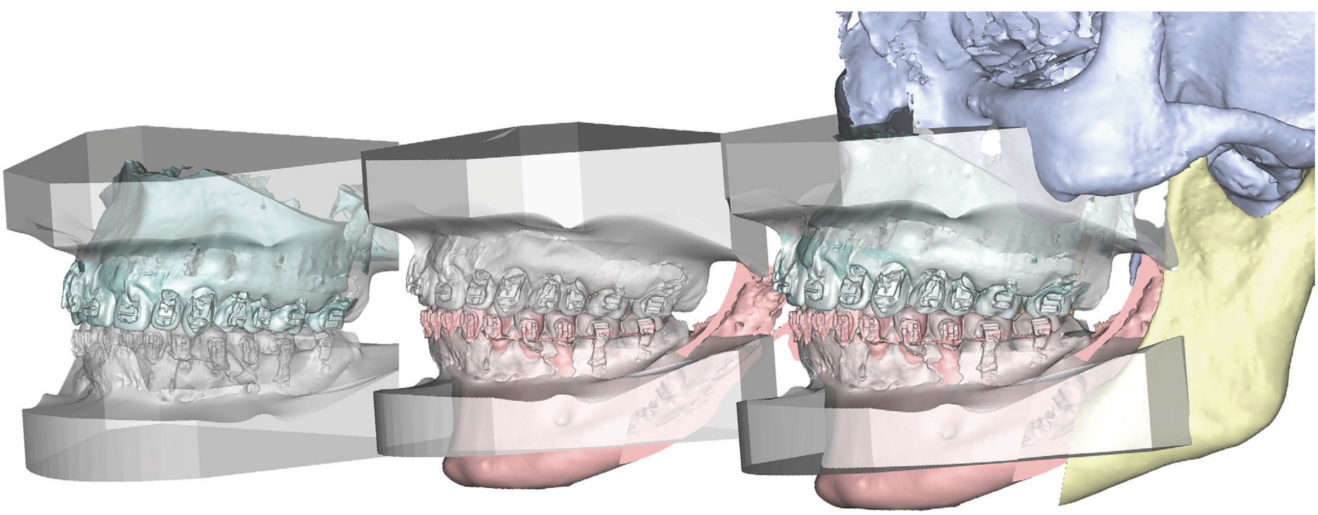


Fig. 11 Sequence of alignment for the maxillo-mandibular complex: left: occlusion aligned to the maxilla (UIM = 0); center: distal segment aligned to occlusion; right: effect of mandibular repositioning with native position of the Le Fort I segment

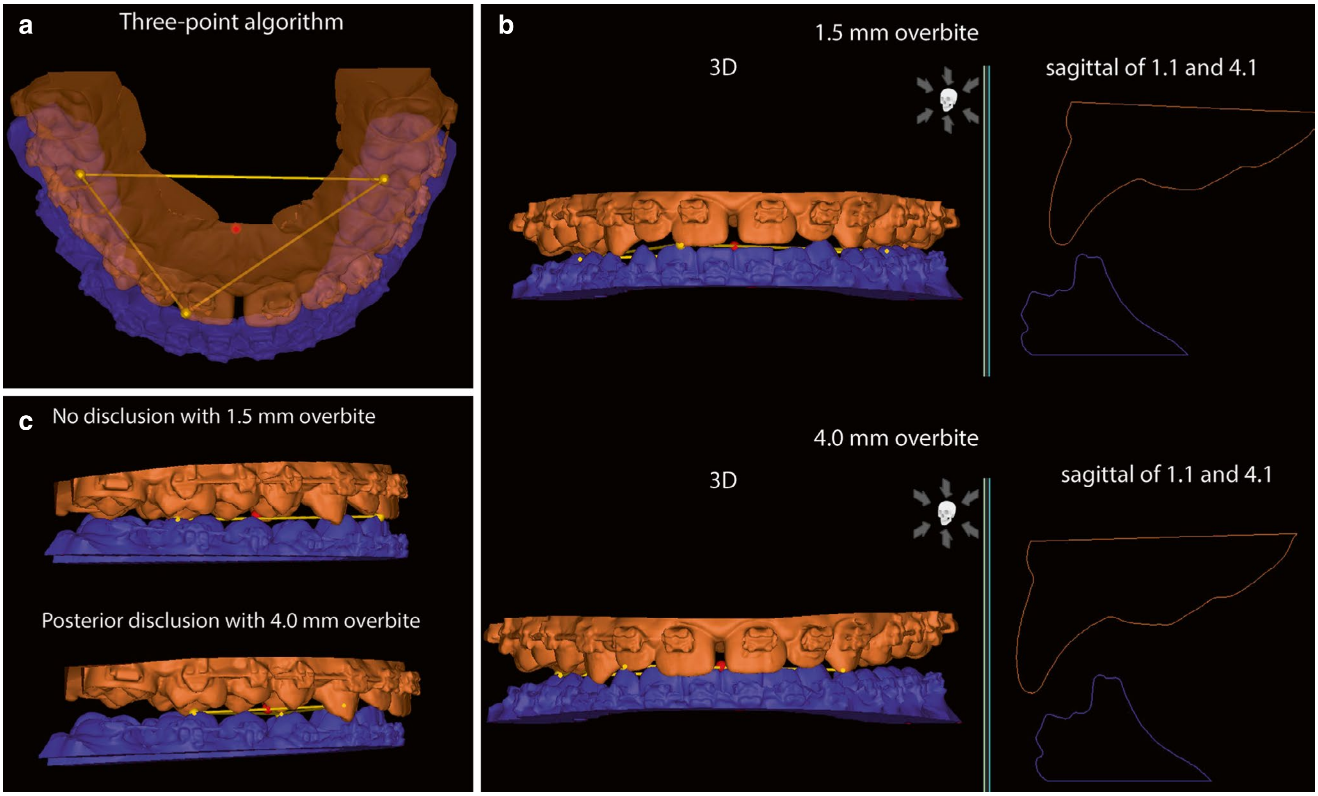


Fig. 12 Virtual occlusion with fine variation of parameters and three-points contact scheme

jet (Fig. 12). Although this method is becoming more robust and validated [9, 10], showing comparable results with the traditional method (Fig. 13), many surgeons are resistant to its adoption. Virtual occlusion provides the most accurate prediction of the final occlusion as it does not involve intermediate scans of alginate impressions and can be immediately applied starting from intraoral scans.

Once the surgical occlusion is determined, both using a scanned or an entirely automated method, it is imported in the virtual surgical plan and is used to register the mandibular distal segment into correct occlusion with the superior antagonist. For the occlusion cast method, the occlusion cast is imported, its superior component is aligned with the Le Fort I segment and is made coincident using a global alignment method; subsequently, its inferior arch provides the template to register the teeth of the mandibular distal segment in correct occlusion. This step is automatically performed by the software when the virtual occlusion algorithm is implemented.

The result of this method is the mandible registered in the correct occlusal relationship with the superior antagonist: a mandibular setback will result in III class, while a mandibular advancement will result in II class. Notably, cephalometric points linked to the mandibular distal segment will now show modifications proportional to the entity of mandible repositioning; for instance, the Δx component of LIM (Lower Incisor Midline) will reflect the entity of overjet correction.

3.8 10-Step Flowchart for Skeletal Repositioning in Orthognathic Virtual Surgical Planning

During surgical movements, the maxillo-mandibular complex in the desired occlusal relationship should be considered a unique entity. Therefore, Le Fort I segment and mandibular distal segment are treated as a single part and not separately, taking as reference cephalometric points applied to the Le Fort I segment that are still in the position $x_0y_0z_0$. From this step onwards, the correction of the dentofacial deformity is planned using a predefined stepwise approach:

1. Cant adjustments (*roll*): an invariable horizontal plane is taken as reference, generally based on the Frankfurt plane (see Sect 3.4). From this reference, perpendicular measurements are drawn to the cusps of canines and first molars, thus measuring their vertical discrepancy that accounts for cant correction. If needed, correction of canting is performed by placing the rotation pivot at the

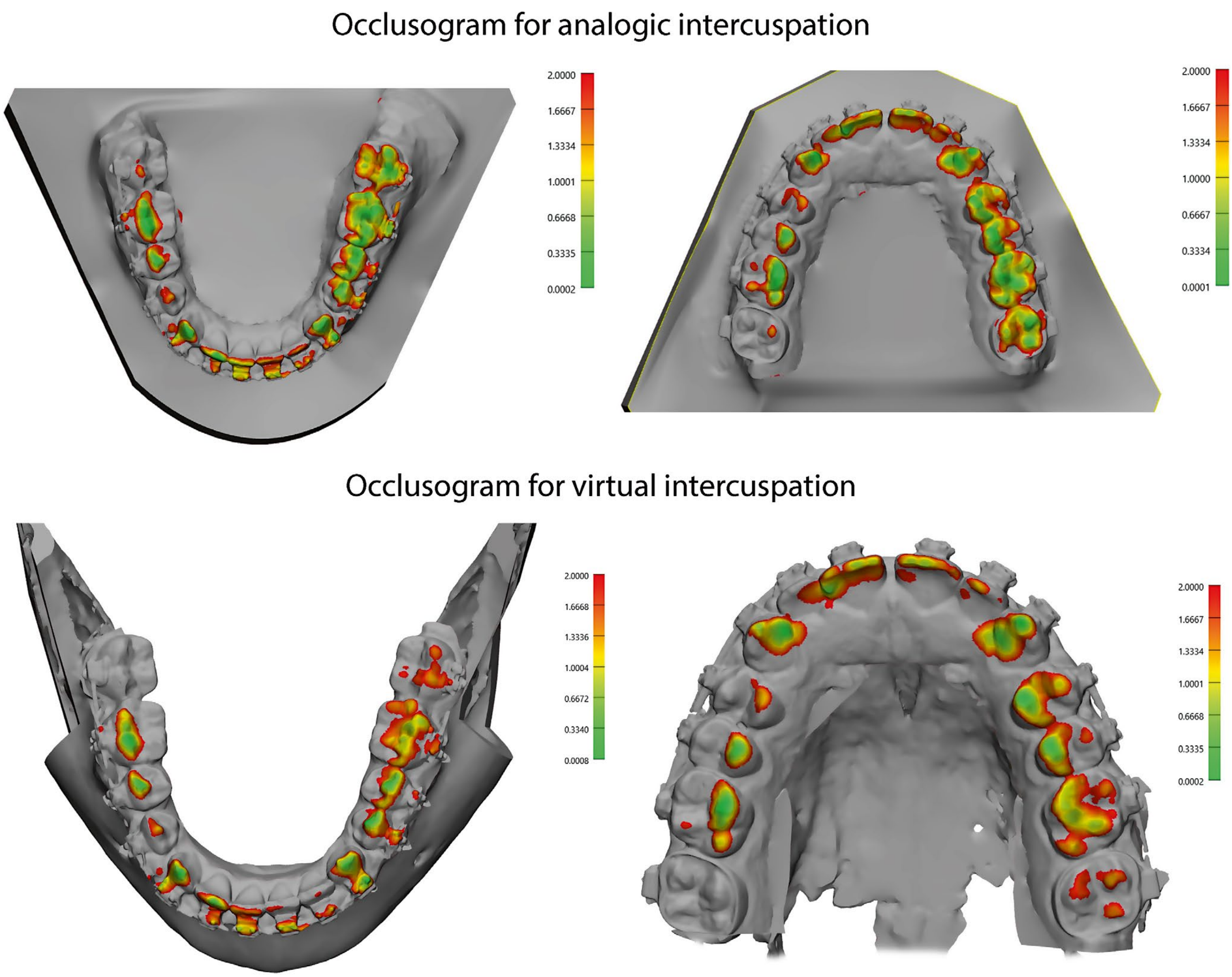


Fig. 13 Occlusograms between analogic and fully virtual intercuspation

lateral extremity of the osteotomy line in the contralateral side. To facilitate this step, the mandible object can be selectively hidden while remaining still selected and affected by correction (Fig. 14).

2. Midline adjustments: we consider as reference midline a vertical line passing through the labialis superior point and subdividing the Cupid's arch into two symmetrical halves. If the superior interincisal line is deviated from the midline, a translation movement is applied to center it. The inferior interincisal line is already aligned with the superior because the mandible is in occlusion (Fig. 15).
3. Anterior-Posterior correction: depending on the profile and dental exposure during smile, as assessed from preoperative clinical analysis, anterior-posterior movements of the maxillo-mandibular complex are performed. This step is crucial to correct maxillary retrusion, as the advancement should be proportional to the increase of volume around the middle third region. Maxillary

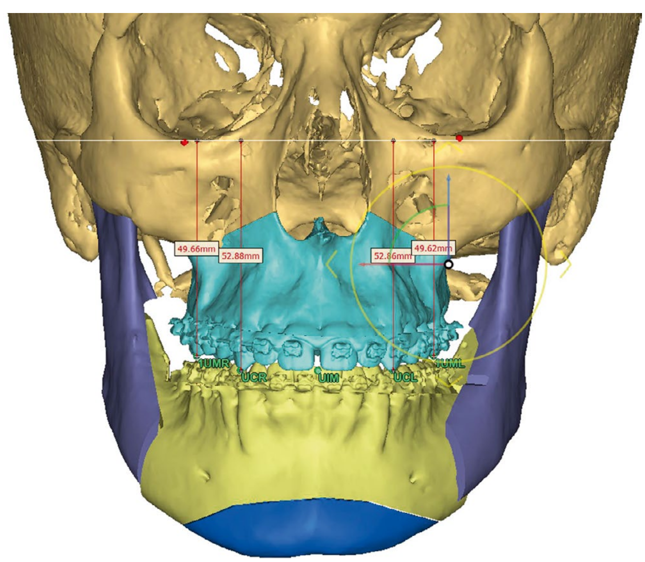


Fig. 14 Evaluation of maxillary coronal symmetry calculated on canines and first molars to ascertain the presence of maxillary cant

advancement also impacts on dental exposure; thus, its amount should be balanced between the correction of a gummy smile (or a reduced crown exposure) and the need for vertical correction (Fig. 16). As mentioned, the mandible is automatically repositioned in its correct occlusal relationship. The main concept of modern computerized VSP is that skeletal repositioning should be based on a comprehensive, holistic, facial analysis and not on the separate evaluation of the upper and lower jaw.

4. Vertical correction: it mostly impacts the dental exposure during smile accounts also for the height of the occlusal plane. Typically, long face phenotype is hyperdivergent, whereas short face phenotype is hypodivergent. Maxillo-mandibular impaction lifts the occlusal plane, decreases the occlusal angle between Frankfurt plane (FHP) and occlusal plane (OcP), and decreases hyperdivergence (Fig. 17). In addition, linear maxillo-mandibular impaction ("in toto"), compared with counterclockwise rotation alone, decreases the amount of lateral gummy smile, acting on the so-called "buccal corridors". Differential maxillo-mandibular impaction refers to the combination of linear impaction with pitch correction.
5. Occlusal plane rotation (*pitch*): this rotational movement acts specifically on the occlusal angle (FHP-OcP). This angle measures on average 8° ± 4° [11]. High occlusal plane phenotypes, which are typically associated with hyperdivergence and increased verticality, require a correction through an anterior rotation (counterclockwise rotation, CCW). Low occlusal plane phenotypes, generally associated with hypodivergence, II class and reduced verticality, require FHP-OcP augmentation through a posterior rotation (clockwise rotation, CW). Figure 18 illustrates the effect of CW and CCW on occlusal plane pitch.

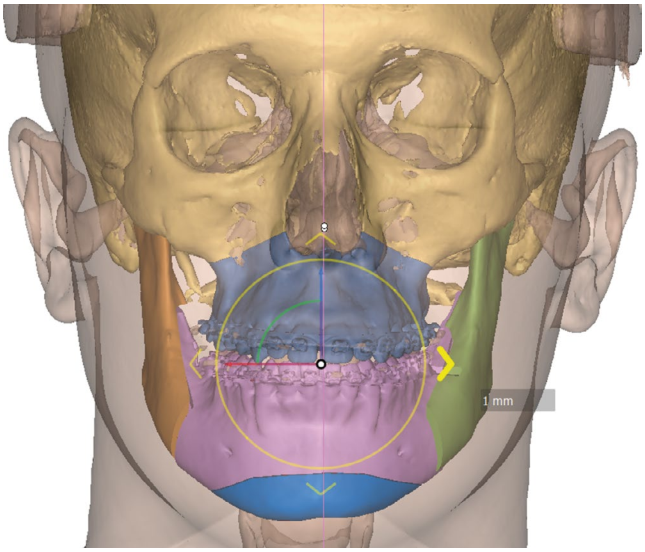


Fig. 15 Midline centering. The soft tissue reference is placed in the midpoint of the columella-Cupid's bow

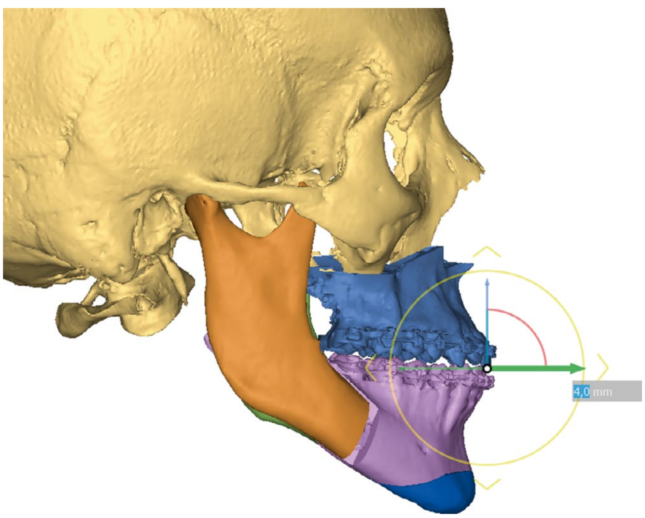
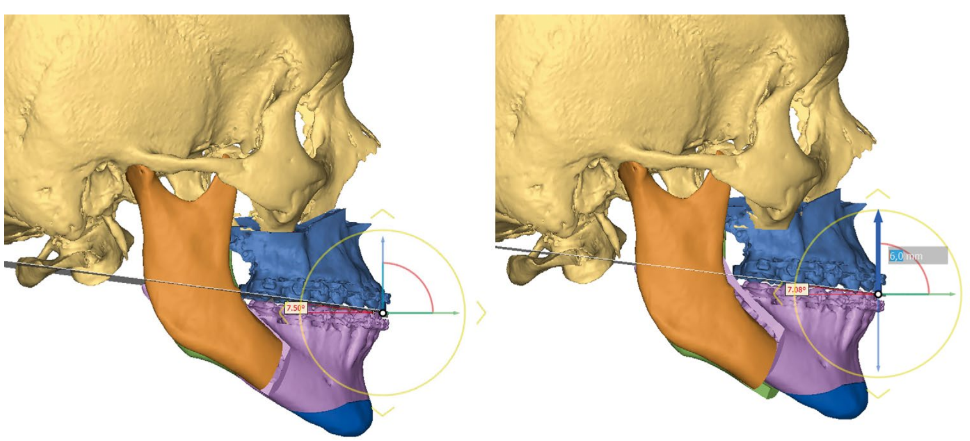


Fig. 16 Sagittal repositioning of the maxillomandibular complex

Fig. 17 Pitch modifications



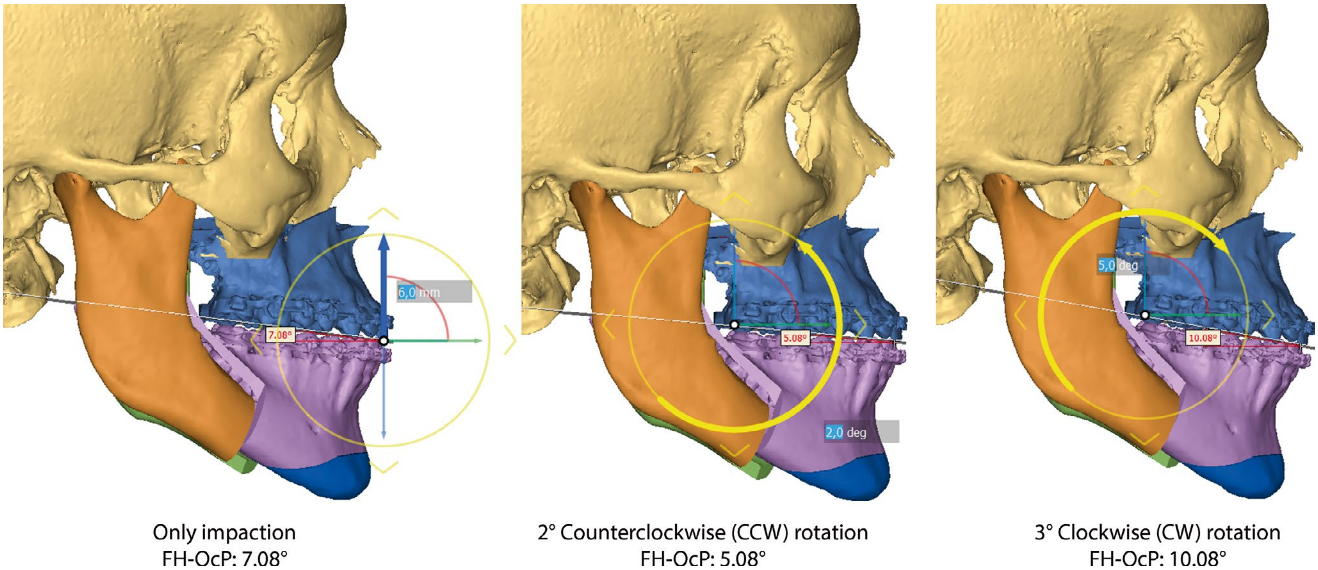


Fig. 18 Effect of CW and CCW on the slope of occlusal plane and the FHP-OcP angle

6. Axial maxillo-mandibular rotation (*yaw*): this movement enables the correction of rotational discrepancies in the axial plane. Axial mismatches must be fist corrected when assessing the natural head orientation. First, a reference grid should be enabled in the software to evaluate whether each tooth lies on the same horizontal line as its contralateral; if not, the axial discrepancy must be corrected to symmetrize. Moreover, this movement is strategical to minimize the bone interference between the proximal and distal segments of the mandible. As this movement affects the interincisal line, already centered, the Author advises to position the rotational fulcrum in correspondence of the interincisal line itself to preserve its alignment with the dental midline (Fig. 19).
7. Genioplasty: the three-dimensional repositioning of the chin completes the VSP in cases where a genioplasty has been planned to further correct the lower third if mandibular repositioning was not sufficient (Fig. 20).

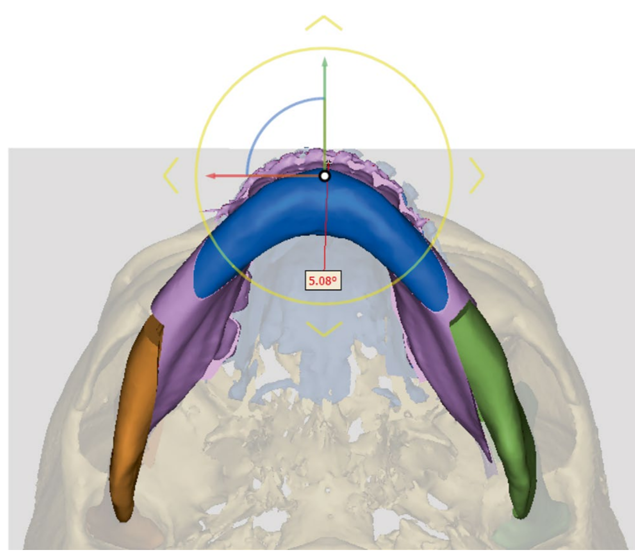


Fig. 19 Yaw correction

8. Assessment of mandibular rami: after placing the rotational pivot in the center of the mandibular condyle, the position of each mandibular ramus is optimized for the new orthognathic setup (Fig. 21). This is particularly important for fully-custom surgery with patient-specific plates. In such specific circumstances, care should be taken to avoid geometrical compenetrating of ramus and distal segment geometries to increase predictability of surgery. If a suboptimal correspondence of mandibular rami with the distal segment is seen, yaw correction should be inspected to ensure that no further improvements are possible. Moreover, excessive correction in the coronal plane should be avoided for its implications in introducing a condylar torque, with detrimental effects on the joint.
9. Visualization of bony interferences and eventual trimming: the assessment of bone intersection is essential to evaluate the necessary ostectomy around the Le Fort I osteotomy to surgically achieve the planned movements. The same is true in case a reduction genioplasty is planned. In addition, it enables to define hindrances between the distal and proximal mandibular segments, which can be solved using a trimming procedure, correspondent to a selective ostectomy during surgery (Fig. 22).
10. Soft tissue simulation: once bone segment repositioning is complete, modern software enables to perform the soft tissue simulation to visualize the effects of skeletal change on soft tissue volumes. Yet this technology has several limitations, it provides a gross idea of the final surgical aspect (Fig. 23). More sophisticated studies in the future will substantially improve this aspect, for instance using finite element analysis (FEA) to regionally determine the resilience of any tissue to the skeletal movement in terms of skin thickness, fat volumes, muscle involvement.

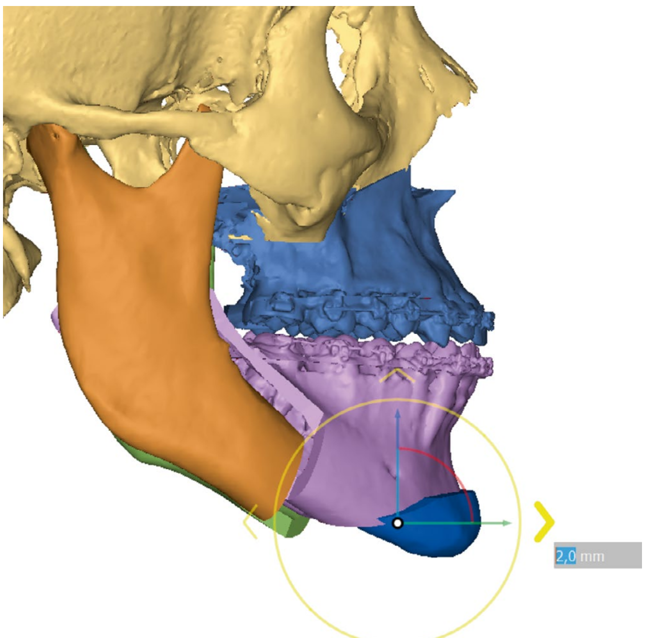
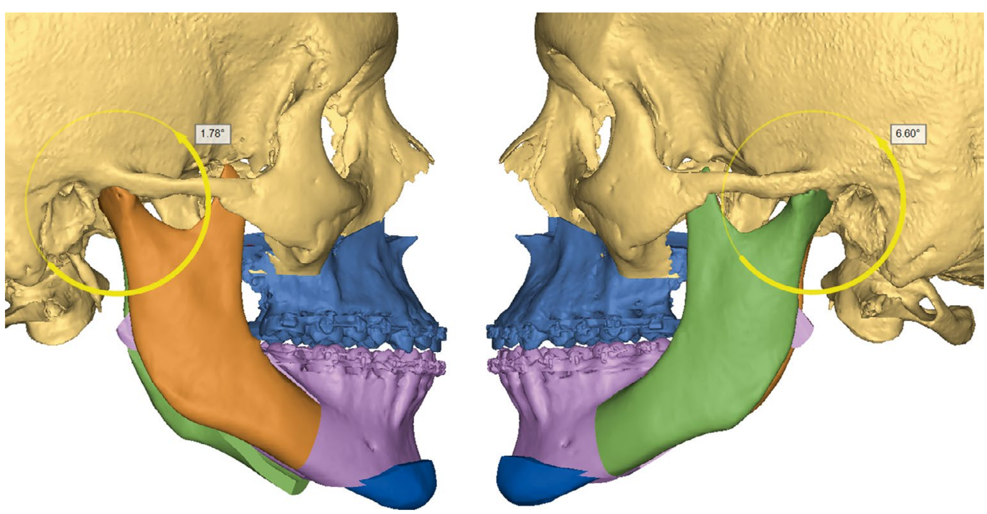


Fig. 20 Genioplasty

3.9 Cephalometry Inspection

After repositioning of the bone segments, it is crucial that each cephalometric points previously placed is carefully inspected to determine its position in relation to the surgical plan (Fig. 24). As mentioned in Sect. 3.5, the most esthetically impactful evaluation is played by the vertical and anterior-posterior values of UIM. Moreover, four angles are crucial as their change is directly related to the corrective orthognathic movements:

Fig. 21 Mandibular rami assessment



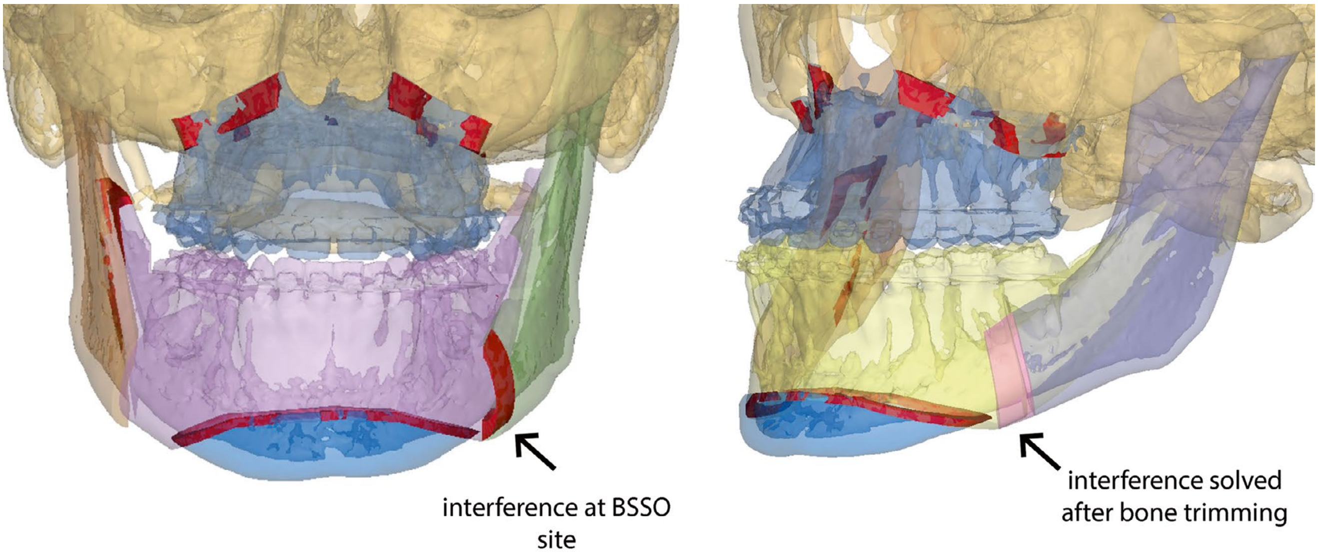


Fig. 22 Evaluation of interferences at osteotomy sites

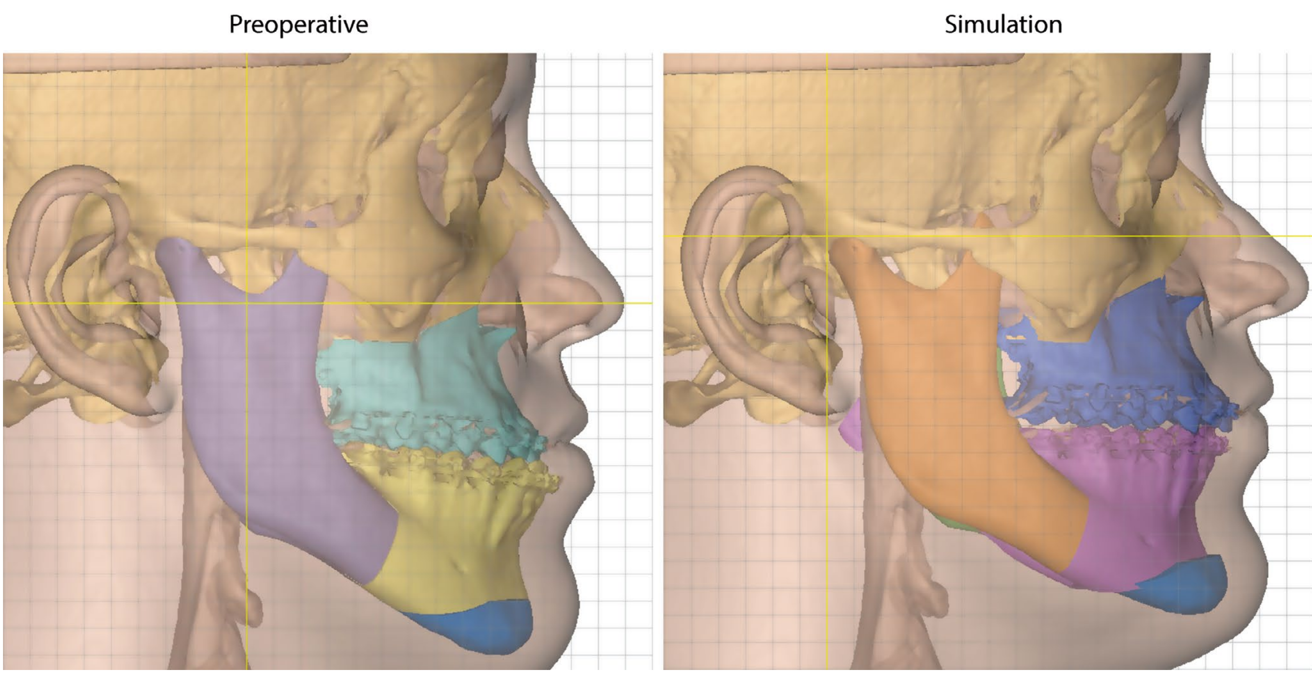


Fig. 23 Simulation of orthognathic surgery on soft tissues

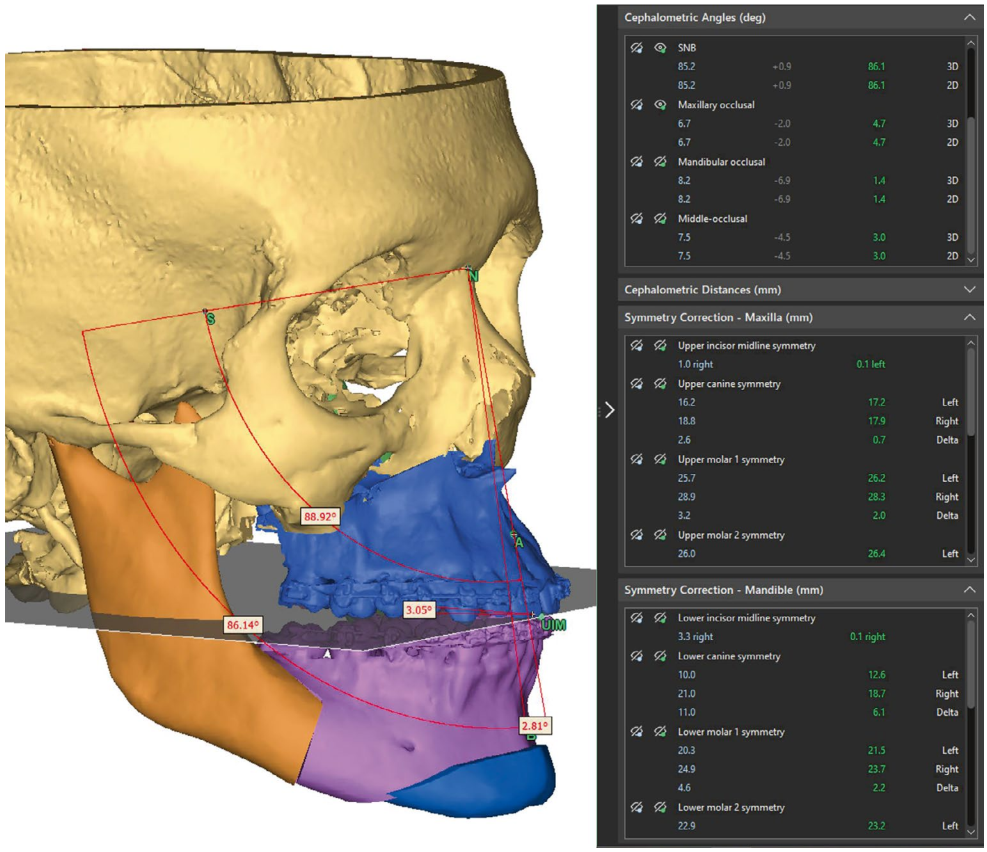


Fig. 24 Cephalometry interactive modification is used to verify the effect of skeletal repositioning across the cephalometric points

- ANB: it defines the entity of the maxillomandibular discrepancy and its slight variations are related with mandible repositioning. Used at first for 2D cephalometric evaluation at its earliest, its implications are less relevant with automatic occlusion. It can be affected and normalized by pitch variations.
- SNA: it defines the sagittal position of the maxilla. Its normalization accounts for the appropriate anterior-posterior repositioning of the upper skeletal base.
- SNB: it represents the SNA lower counterpart.
- Occlusal plane (FHP-OcP): its correction is particularly important when related to the high/low occlusal plane phenotypes, as described in Sect 3.8.

3.10 Ancillary Procedures

Any additional procedure, including mandible contouring ostectomies or malar augmentation, should be planned on the final orthognathic setup. The software already provides the appropriate tools to perform such basic osteotomies and there is no need to shift to other more advanced software.

3.11 Measurements

Once the final plan is validated, is important to produce a report with the following data:

- Planned cephalometry, displaying the ($\Delta x,y,z$) for each cephalometric points related to skeletal movements.
- Maxillary bone intersection quantification: in the absence of a surgical guide, not routinely used for orthognathic surgery, it is important that, in case of a maxillary impaction movement, the surgeon knows the exact quantity of bone to excise to achieve the desired movement. This is even more important when a pitch or roll correction is planned owing to a differential osteotomy design (the ostectomy differs between posterior and anterior, and/or left and right). Measurements are calculated on the bone interference model and taken lateral to the piriform notch and the zygomatic buttresses bilaterally.
- Similarly, it is important to know the amount of bone to excise in the proximal mandibular segment (the ramus) to avoid undesired interferences with the proximal segment which would hinder osteosynthesis. This quantity of bone

is optimally determined during trimming of interferences (see Sect. 3.8, point 8).

- If a reduction genioplasty is planned, likewise the surgeon has to know the exact quantity of bone to excise.
- Maxillary advancement: especially in case of combined, nonlinear movement, including not only translations but a pitch and roll correction, a three-dimensional measurement should be conducted to quantify the magnitude of the "step" between the advanced Le Fort I segment and the skull model.

3.12 Choosing the Surgical Sequence (Intermediate Position) and Designing Splints

Depending on the clinical case, there are two possible surgical sequences [12]:

- Maxilla first: in the splint creation phase, the intermediate position is defined by the final position of Le Fort I segment (upper maxillary) and the preoperative position of the mandibular distal segment (lower maxillary). Especially in second classes with marked overjet which require a simultaneous maxillary advancement, the maxillary-first intermediate position might lead to the creation of a surgical splint with very distant dental arches, thus hindering the application of metal wires for intermaxillary block creation. Such cases might be strategically approached with a mandibular first sequence to facilitate intermaxillary block (Fig. 25a).
- Mandible first: in the splint creation phase, the intermediate position is defined by the final position of the inferior maxillary and the preoperative position of the superior maxillary. This choice, yet preferrable for several aspects [13], raises some concerns during the virtual surgical planning in case a maxillary impaction is planned, because this movement causes an upward displacement of the occlusal plane. As a consequence, the preoperative maxilla will be lower than the final occlusal plane of the mandibular distal segment, and the two will compenetrate each other. The solution is to simulate ad additional disclusion of the final mandible, placing the rotation fulcrum over the TMJ and increasing the rotation of the mandible distal segment to avoid the compenetration with the preoperative upper maxillary, making the creation of the splint possible [14]. This maneuver, however, introduces a slight inaccuracy, even if correctly performed, because it is based on a fictitious movement, as a normal TMJ exhibits a complex combined rotation and translation movement. For small rotations, however, the error can be considered negligible (Fig. 25b).

Based on such principles, splint design can be performed entirely within the orthognathic surgical planning software through a dedicated module by placing marker points across the upper and lower dental arches. Points should be placed lower than the center of brackets to avoid bracket interference; in other words, the negative impression on the splint should take into account the volume of the bracket. Last, it is of prominent importance that undercut removal is applied before the creation of any splint.

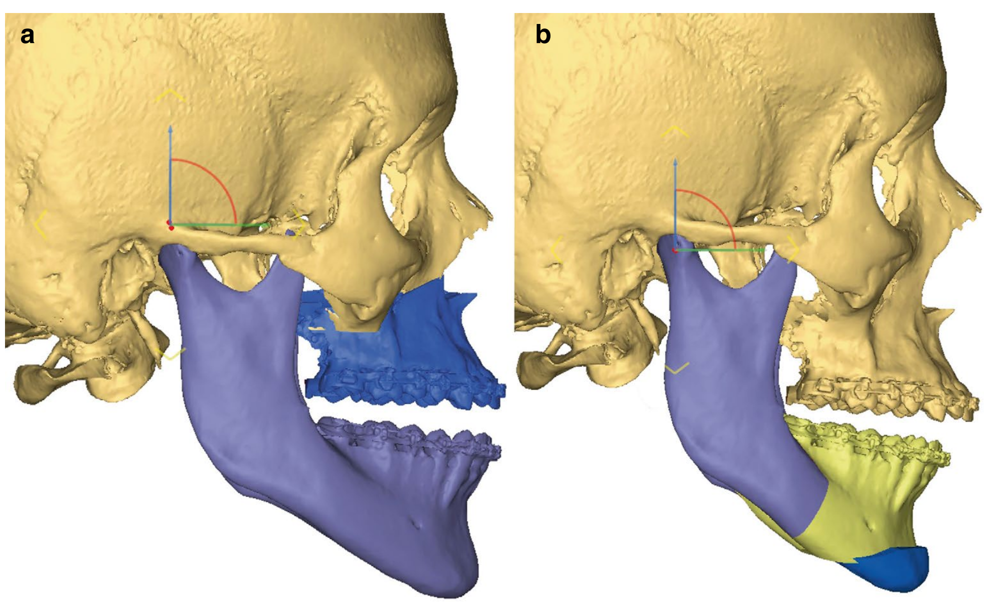


Fig. 25 Establishing the surgical sequence: maxilla first (left) or mandible first (right)

4 Sequences of Phases for Bimaxillary Orthognathic Surgery with Multipiece Maxillary Segmentation

Multipiece orthognathic surgery is generally required to correct transversal defects or severe discrepancies of the Spee curve, which can not be compensated exclusively orthodontically. Compared with a two-stage process that involves surgically assisted rapid palatal expansion (SARPE), two-piece or three-piece maxillary segmentation has the benefit of addressing severe transversal defects in the same surgical time and is particularly indicated in a variety of situations, including posterior crossbite and anterior open bite. Moreover, three-segment maxilla has several additional advantages over SARPE, including avoidance of maxillary fibrosis and scarring due to a previous operation; less esthetic impact, because the space created by the expansion is divided between right and left interproximal spaces between laterals and canines (or canines and premolars) instead of causing a disfiguring midline diastema; prevention of damage to the midline dental papilla; preservation of the nasopalatine bundle; and possibility for an asymmetric expansion [15].

Segmentation of the maxilla in orthognathic surgery introduces several elements of complexity compared to the workflow described for basic bimaxillary orthognathic surgery. For multipiece maxillary surgery, many surgeons still prefer the physical setup performed on plaster models and articulator ("model surgery"). However, virtual occlusion for multipiece maxillary surgery is providing interesting clues for a faster, fully-digital, workflow, which can eliminate the tedious and costly processes of model surgery.

4.1 Traditional Model Surgery

Analogic setup often requires the availability of a dental technician, and involves cutting plaster models with a saw, with possible damages to the models themselves. Cut maxillary segments are positioned in the optimal setup within the desired occlusal relationship using the articulator, and separate pieces are then sticked together using hot glue or wax (Fig. 26). The setup needs to be rescanned to provide the final occlusion for the VSP, thus adding another analogical-to-digital transition, which might introduce further inaccuracies. Both pre-setup, native, dental arches scans and post-setup occlusion need to be provided to the software.

4.2 Osteotomies for Maxillary Partition

Osteotomies for maxillary partition have to be simulated on the native maxilla with the preoperative intraoral scan aligned and superimposed, cut through the Le Fort I plane. Given the importance of performing such osteotomies without violating teeth apices, it is of prominent importance to visualize a volume rendering of dental roots when positioning cutting planes for segmental osteotomies. Moreover, the orthodontist should have created a gap when the osteotomy is planned to decrease this risk. For a three-piece partition setup, virtual osteotomies can be performed using two design subtypes (Fig. 27):

- A "Y-shaped" cut is the easiest to perform using planar cuts and yield a configuration much similar to that of the model surgery, allowing to compare the virtual setup to the physical one in the surgical planning.
- On the contrary, a "central island" design is more difficult to achieve, but it mirrors the intraoperative expansion, allowing to reproduce a configuration to assess the intraoperative situation.

The same principles apply to maxillary bipartition where both a single midplane splitting the maxilla in two symmetric halves and a midplane crossing the H-shaped design can be simulated.

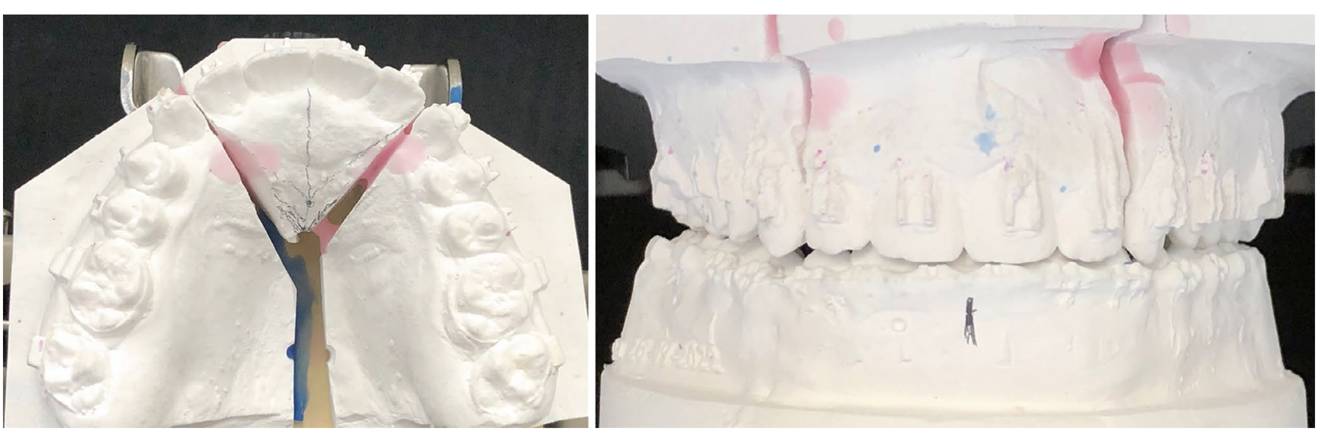


Fig. 26 Model surgery performed on plaster models to simulate three-piece maxillary partition

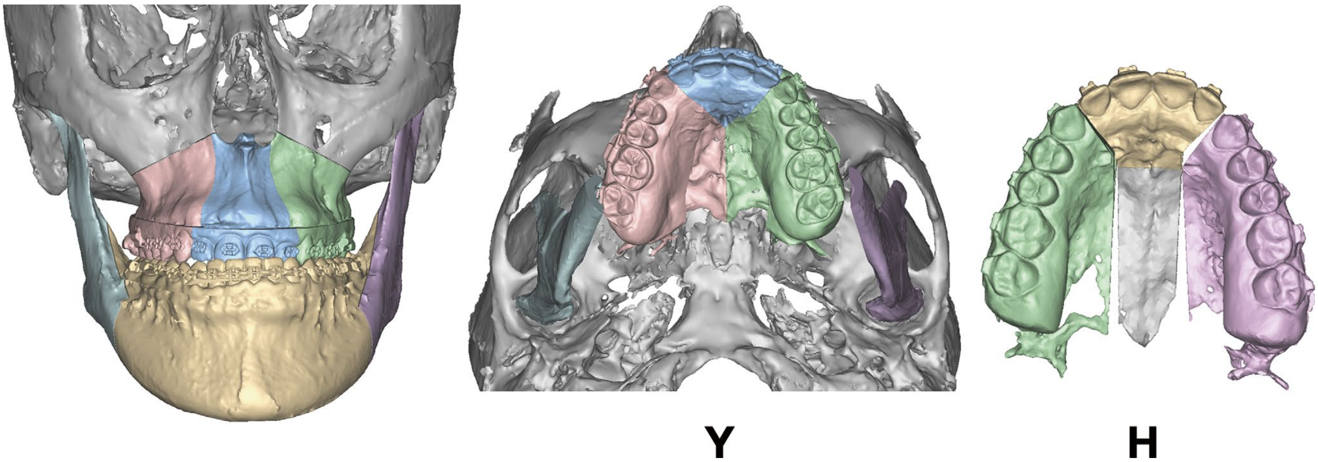


Fig. 27 Comparison of two design methods for maxillary tripartition in VSP: Y-shaped (more practical), or H-shaped (mimics what happens during surgery)

4.3 Virtual Multipiece Occlusion

Virtual multipiece occlusion is a new and highly promising frontier. It involves a complete, digital, setup, avoiding the time and resources spent in performing the analogical articulation of sawed plaster models. Moreover, it works using exclusively native intraoral scans, and the quality is not worsened by subsequent, additional rescans of analogical setup. After segmental osteotomies have been simulated in the VSP, left and right hemimaxillary parts (for bipartite setup) or left, right posterior sectors and premaxilla (for tripartition) can be handled by the software, which detects the most suitable configuration. In contrast with automatic setup for a full-arch maxilla, for segments the algorithm can adopt maximum two contact points, where the user is allowed to modify coefficients that include buccal-lingual translation, mesial-distal translation, axial rotation and buccal-lingual tilt (for lateral segments), or midline deviation, overjet, axial rotation and sagittal rotation (overbite) for the premaxilla. This approach holds great value as it allows to predict if a conventional bimaxillary surgery, planned as such, could benefit from a multipiece setup, in the eventuality that this was not considered both from the surgeon and the orthodontist. The surgeon can make countless simulations in the VSP to find the best arrangements and understand whether a single arch preparation is sufficient or if a multipiece maxillary partition is needed, both in the presurgical VSP, as well as during the orthodontic preparation, which can drive the orthodontist to shift toward a preparation for multipiece osteotomies by widening interdental spaces where osteotomies are planned (Fig. 28).

4.4 Transferring the Physical Setup in the VSP (Not Needed for Virtual Occlusion)

Subsequently, the setup model is imported in the virtual planning and is used as template to rearrange the virtual segments exactly as the physical setup. In a three-piece maxillary surgery, the alignment chain proceeds as follows:

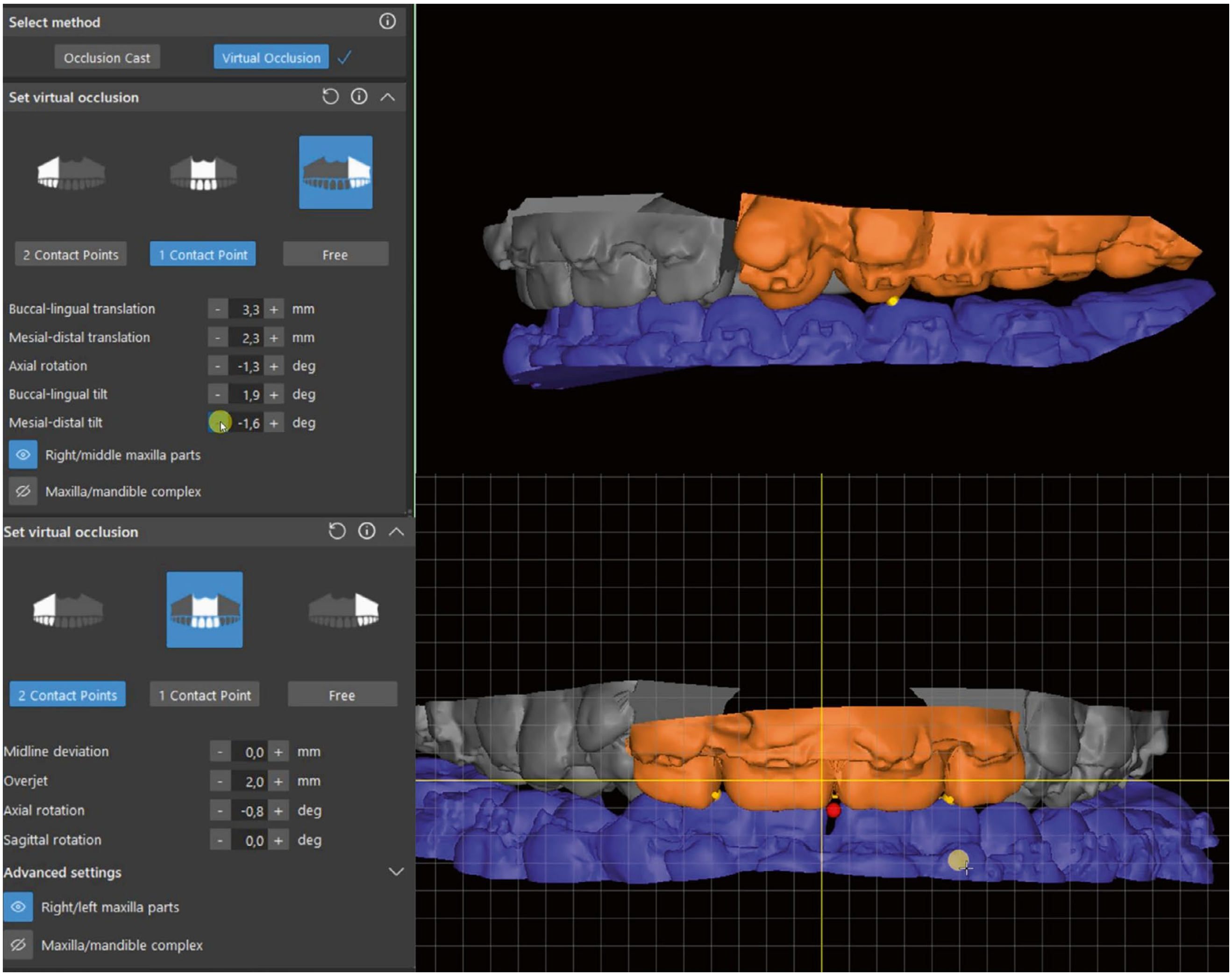


Fig. 28 Virtual occlusion applied to multipiece maxillary setup. Each individual segment is managed separately, allowing to correct Spee curve and transverse diameter

- First, the premaxilla of the model setup (moving) is aligned to the premaxilla of the native patient (fixed) in order not to alter the sagittal position of the superior incisor, which remains in the position $x_0y_0z_0$. The superior incisal point (UIM) is the most important parameter to monitor when assessing surgical movements, because it has a direct implication on the patient's smile. This step ensures that premaxilla is in a neutral position, and subsequent registration phases will define the transversal expansion of the lateral segments, which are taken into account also in the cephalometric points in the molars.
- Second, once the premaxilla of the plaster model setup is coincident with that of its virtual homologue, both lateral sectors are aligned with the molars of the analogic setup. This step enables the correct expansion of the maxilla exactly as planned in plaster models (Fig. 29).

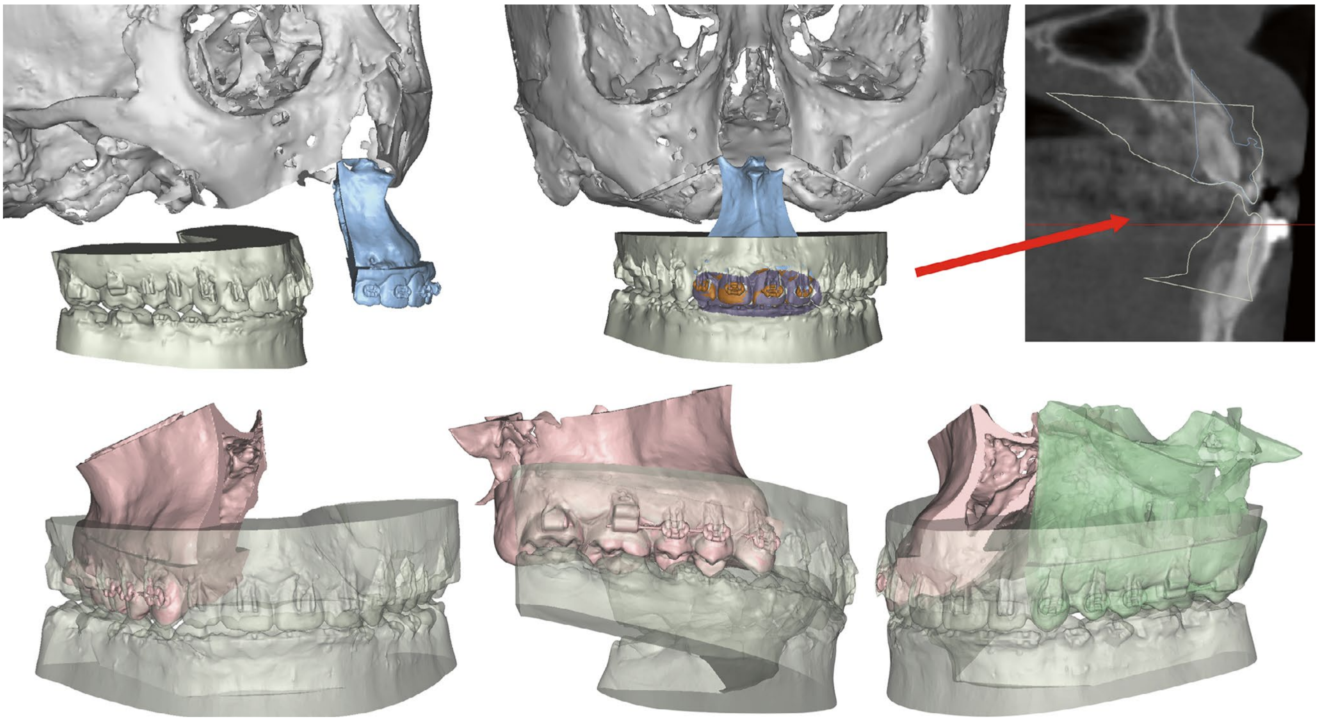


Fig. 29 Multipiece maxilla VSP using plaster model setup. The first step is alignment of the setup with the premaxilla, maintaining UIM at 0 value. Then, lateral segments are expanded according to the setup

- Verification of the plaster model setup with the expansion achieved in the virtual surgical plan: this step is essential to assess that dental cusps of the expanded segments match the cusps of the imported setup. For Y-based osteotomy design in tripartitions and for the simple midplane in bipartitions, the movement of bone segments can be specular and more intuitive to demonstrate (Fig. 30).
- Following the scanned plaster model setup, the maxillo-mandibular occlusion is established by repositioning the mandible according to the model surgery occlusion. At this point, the maxillo-mandibular complex in the correct occlusal relationship can be repositioned to achieve required facial movements and the desirable occlusal plane tilting (Fig. 31).

In case of maxillary splitting across the midline, the procedure is the same, with the difference that there are only two segments to realign rather than three. The model surgery setup will be realigned to the less expanded segments and, once in position, will drive the contralateral half in the correct occlusal relationship.

From the analysis of both workflows, fully virtual and half-analogic with model surgery, it is clear that the fully virtual workflow excels in terms of intuitiveness and decreased number of steps, which translates into decreased possibility of introducing errors.

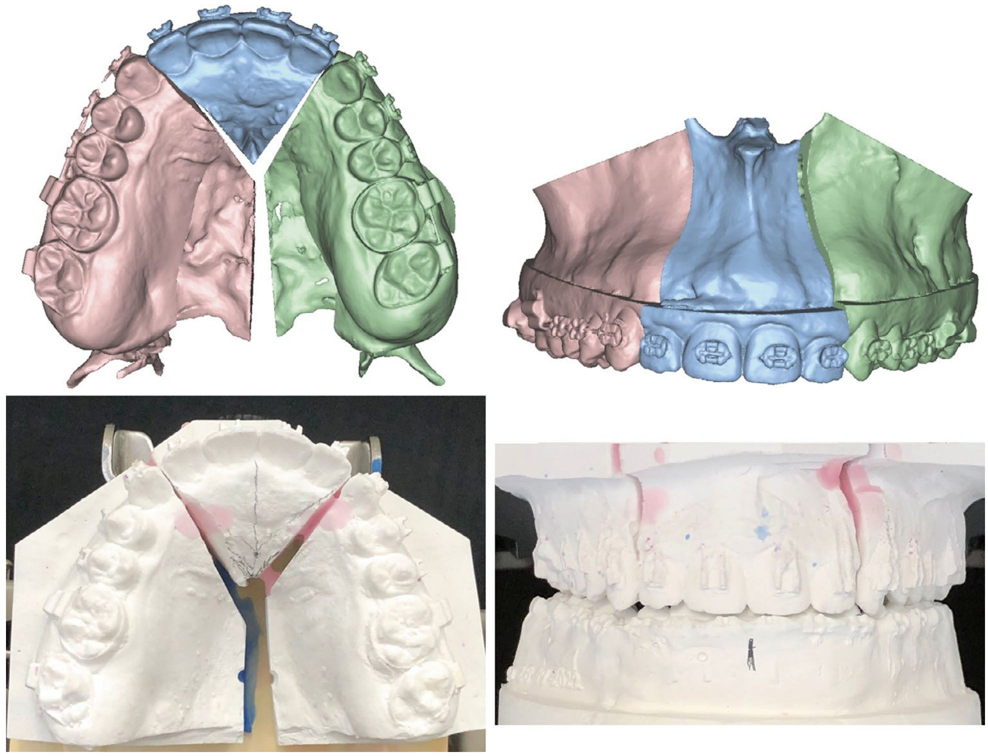


Fig. 30 Expected modifications in plaster model surgery are reliably mirrored by VSP alignment, confirming the correct expansion of the Le Fort I segment

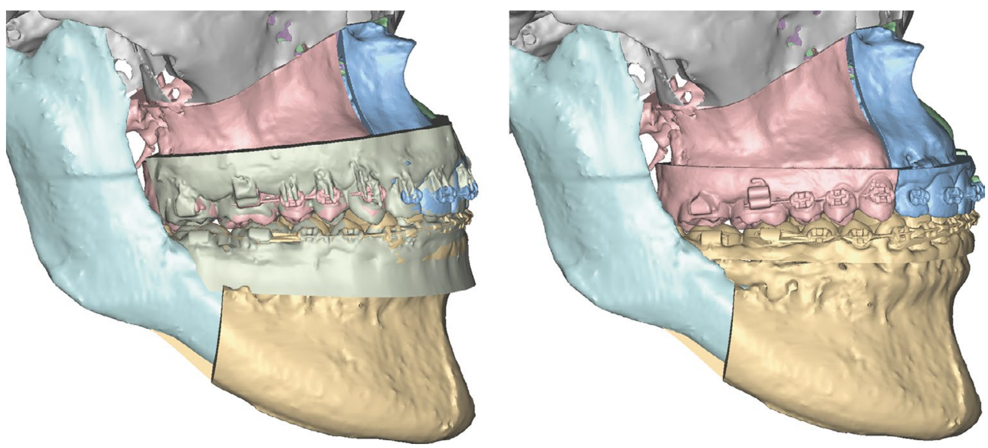


Fig. 31 Mandibular repositioning in relation to the expanded Le Fort I

5 Fully Custom Orthognathic Surgery

Fully custom orthognathic surgery refers to the process of establishing a VSP, as seen so far, and designing patient-specific surgical guides for computer-guided osteotomies, as well as personalized osteosynthesis plates [16]. This process aims at creating automation in surgery, decreasing variability and achieving an overall more standardized and accurate workflow according to the preoperative virtual plan. As the final osteosynthesis plates are designed on the provisional, final configuration after skeletal repositioning, they already incorporate the necessary skeletal movements, allowing to discard the use of traditional dental splints. For this reason, fully custom orthognathic surgery is also referred to as "splintless" surgery.

The fact that surgical guides allow to predrill holes in the correct final positions and that plates bridge skeletal segments in the planned arrangement overcomes the traditional sequencing (maxilla or mandible first) as well. The sequence is fully interchangeable based on surgeon's preferences, as it does not rely upon a specific splint [17].

Plates are designed as wrapping envelopes around the final skeletal position, after defining the most appropriate sites for screw insertion, which can be placed based on considerations of bone wall-thickness, anatomical variations and accessibility. Once screws are placed, the final skeletal configuration, along with the screws themselves, is brought back

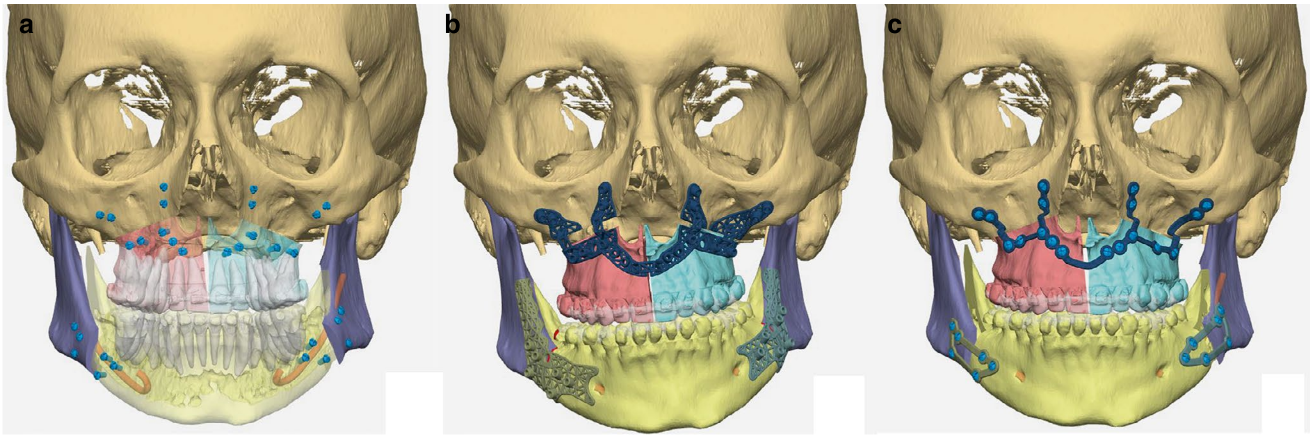


Fig. 32 Sequence of planning in fully custom orthognathic surgery. (**a**) Screw placement; (**b**) realignment of screws with the initial position and creation of the surgical guides; (**c**) final custom plates

to the preoperative configurations, where the same holes are used to fix surgical guides. Cutting planes used to simulate osteotomies in the VSP are taken as reference to define the cutting path within the surgical guide (Fig. 32).

5.1 Workflow for Fully Custom Orthognathic Surgery

The workflow for fully custom orthognathic surgery necessarily involves a medical device company that has the technology to manufacture customized plates, which are implantable devices. There are several steps of increasing complexity that can be used:

- Outsourcing to the external company both the VSP and the manufacturing: it is the oldest and less proficient workflow. It is the preferred choice for centers that do not have the clinical and technological expertise to perform the VSP. As such software is widely available and consistently clinically friendly, the traditional "on-call" VSP performed in teleconference, where the company performs the VSP under the surgeon's supervision, is declining.
- Clinicians performing VSP outsource the final plan to the company for manufacturing (hybrid workflow): it represents the most robust workflow. Using VSP software, surgeons are able to plan the case by themselves, and, once the plan is complete, they deliver it to the company for splint creation. Intermediate stages may involve additional meetings to establish if the position of the screws is correct and solve other issues that may arise. However, this workflow fastens the whole process and is not prone to misunderstandings in the VSP.
- Clinicians performing VSP and designing surgical guides, the company manufactures the plate (advanced hybrid workflow): it is the most advanced stage. For clinicians wishing the highest control and having the right expertise, not just the VSP is performed in hospital, but also the design and production of the surgical guides. In some cases, experienced surgeons might wish to draft plates as well and share their design with the manufacturer, which modifies it according to the regulation criteria established in the company. This has utility especially in case surgeons wish to experiment design variations or prototypes for fixation plates or other implants.

5.2 Precautions in VSP for Fully Custom Surgery

The most impactful maneuver in orthognathic VSP for fully custom procedures is the assessment of bone interferences. This is true in particular for the mandible: after the maxillo-mandibular complex in occlusion has been repositioned, mandibular rami be still in their preoperative position. The final position requires that rami are placed in the most optimal arrangement with the repositioned distal segment by placing their fulcrum of rotation in the center of the mandibular condyle. Any compenetration between the repositioned distal segment and mandibular rami should be minimized by accurately addressing yaw correction, if possible. As the BSSO customized plates are designed in the final position, the final VSP must clearly locate and indicate areas of bone interference to be removed, avoiding to fixate customized plates against a tension caused by posterior inferences.

6 AI-Assisted Digital Workflow in Orthognathic Surgery

The full-digital workflow in virtual surgical planning (VSP) of orthognathic surgery requires meticulous planning to achieve satisfactory clinical outcomes. Various steps in VSP, such as segmentation of anatomic structures, rendering of 3D augmented head models, cephalometry, establishment of the final occlusion, and determining optimal virtual jaw displacements, involve multiple, complex manual operations that are time-consuming and prone to operator-dependent errors. In recent years, the integration of artificial intelligence (AI) into VSP is enhancing the accuracy and efficiency of this process, providing surgeons with advanced (semi) automated tools for image analysis, predictive modeling, and treatment simulations.

6.1 AI-Assisted Segmentation and Rendering of 3D Augmented Models

Recent developments in deep learning have facilitated the traditionally labor-intensive identification and segmentation of various anatomical structures that are required for VSP in orthognathic surgery, such as teeth, jaws, the temporomandibular joint (TMJ), and the mandibular canal.

Automated segmentation of jaws from CBCT scans has been achieved using a mixed-scale dense (MS-D) convolutional neural network. In this way, simultaneous segmentation of the jaw and teeth can be carried out accurately, within approximately 25 s for each CBCT scan [18]. By using convolutional neural network (CNN), combining PointCNN and 3D U-net, teeth on intraoral scans (IOSs) can be segmented and labelled automatically, with clinically acceptable accuracy [19]. The segmented and labeled teeth data from IOS can subsequently be superimposed on the segmented jaws from CBCT through AI-driven registration. A recent randomized clinical study has demonstrated that in the absence of imaging artifacts, the accuracy of this multimodal AI-driven registration is similar to conventional superimposition methods [20]. By combining different AI-based approaches, a 3D augmented head model with accurate image data of the dentition, jaws and facial soft tissues can be created, upon which VSP can be performed.

6.2 AI-Assisted Diagnostics in VSP

Conventional presurgical diagnostics in orthognathic surgery include hard tissue and soft tissue cephalometry. Furthermore, it is important to evaluate the morphology and seating of the condyles to determine the appropriate sequencing of double-jaw surgery and to assess the risk of postoperative condylar resorption and skeletal relapse. Accurate assessment of the location of the mandibular canal with regard to the mandibular buccal cortex and lower border is important to minimize the risk of direct injury to the alveolar nerve during osteotomies. Latest advancements in AI-assisted techniques have improved all these diagnostic steps in VSP.

AI-assisted cephalometry uses deep learning models to automatically identify cephalometric landmarks on CBCT images and perform automated linear and angular measurements. CEFBOT is a deep learning-based method, which applied convolutional neural networks (CNNs), to identify skeletal and soft tissue landmarks on 2D cephalograms. The accuracy of Arnett's cephalometric measurements obtained by CEFBOT was comparable to that of experienced clinicians [21]. By using deep learning approaches, automated systems for cephalometric landmarking on 3D CBCT images can also be achieved with a mean localization error of 1.0 mm, with a reliability for skeletal measurements on par with what clinicians obtain. However, dentoalveolar measurements are less accurate, though satisfactory for hard and soft tissue cephalometry in VSP [22]. .

In assessing the condyles' morphology and seating, deep learning-based image segmentation methods have demonstrated high accuracy in segmenting the TMJ, both the condyles and glenoid fossa from CBCT images [23]. Subsequently, by using integration of multistep segmentation, ray-casting and a feed-forward neural network, an automated evaluation of the condylar seating can be performed in VSP [24]. In this way, incorrectly seated condyles can be identified so that adequate actions can be taken by the surgeon prior to surgery to ensure predictable outcome, such as re-scanning of the patient or adopting a mandible-first sequencing in bimaxillary surgery.

Additionally, the location of the mandibular canal, important for a safe osteotomy, can be identified automatically using AI-driven image analysis. A recent study presented a robust and clinically applicable AI system capable of fully automated segmentation of the mandibular canal on CBCT scans obtained from multiple centers [25]. Compression of the main trunk of the mandibular canal by third molar roots can even be visualized in 3D. This high accuracy supports precise surgical planning and helps avoid injury to the alveolar nerve during surgery.

6.3 AI-Assisted Virtual Surgical Planning

In conventional 3D VSP, virtual osteotomy lines were constructed on the augmented head model after which the planned jaw movements are performed in the virtual environment based on the final occlusion, cephalometry and soft tissue simulation of the postoperative outcome. The challenge is to determine the optimal jaw movements required for the patient to achieve the desired occlusion and facial appearance. Advancements in AI are assisting clinicians to accurately simulate postoperative facial changes and to determine the most appropriate jaw positioning.

One of the first published studies in AI-assisted soft tissue prediction used a deep learning-based algorithm for soft tissue prediction in cases of BSSO advancement [26]. Despite being the first study, a higher accuracy was achieved compared to the mass tensor model used in IPS CaseDesigner (KLS Martin, Tuttlingen, Germany), with a mean absolute error for the lower face region of 1.0 ± 0.6 mm. This demonstrated the potential of deep learning to improve the accuracy of soft tissue simulations compared to conventional biomechanical models.

Recently, the facial shape change network (FSC-Net), a geometric deep learning framework designed to predict facial shape changes based on bony movements, was introduced [27]. This network operates by deforming the preoperative facial shape to the predicted postoperative outcome, while simultaneously learning the complex relationship between bone and facial tissue transformations. FSC-Net achieved a 15× increase in simulation time compared to finite element modeling methods and maintained a comparable high level of accuracy. This approach provides surgeons with a more efficient tool for predicting postoperative facial appearances, assisting them to optimize jaw movements during VSP.

To be able to perform soft tissue-based VSP, there is a necessity to be able to automatically determine the optimal jaw displacements required to achieve the desire soft tissue profile. To meet this requirement, a bidirectional 3D deep learning model known as P2P-ConvGC was developed [28]. This model not only can predict the facial soft tissue profile from manually displaced jaws; its strength is that it can also determine the specific bone movements necessary to obtain the desired soft tissue outcome. This innovation represents a significant advancement in automated soft tissue-based backward planning and is bringing orthognathic surgery closer to a more personalized surgical planning.

6.4 Challenges and Future Perspectives

In summary, AI techniques have brought significant improvements to orthognathic surgery by enhancing the accuracy and efficiency of VSP. Key processes like the segmentation of anatomical structure, cephalometric analyses and predictive modeling, which conventionally relied on extensive manual input and sophisticated biomechanical and statistical modeling, have been made faster and less observer-dependent. As a result, surgeons can focus more on critical aspects of patient care, enhancing the overall efficiency, quality and consistency in the orthognathic healthcare pathway.

Despite its transformative potential, AI-assisted VSP faces several challenges. Current AI models still struggle to fully understand the complex relationship between facial soft tissue, underlying skeletal structures and dentition. The calculation of the best-fit final occlusion, which is a key step in VSP, remains beyond the capabilities of AI to date. Furthermore, the development of robust AI models depends on the availability of large, diverse and complete datasets, which are difficult to obtain between clinics and countries. Moreover, there is a lack of a standardized validation method, which is necessary to compare the effectiveness of different AI models in orthognathic surgery in a uniform manner. Additionally, aspects such as privacy concerns, ethical considerations and the integration of AI tools into existing software and clinical workflows remain to be obstacles in the widespread adoption of AI-assisted VSP.

Looking forward, the future of AI in orthognathic surgery is bright. The integration of AI into robotic-assisted surgeries may further enhance surgical precision. With the potential to provide real-time guidance during surgery, autonomous treatment algorithms could allow surgeons to focus on more complex decision-making while AI handles routine, yet critical, tasks. The development of more generalized AI models that mimic the surgeon's thought process in initial treatment planning is also underway. This has the potential to enable AI to adopt a more holistic approach to surgical planning, making orthognathic surgery more personalized and tailored to the unique anatomical features of each patient, while still maintaining the human touch in patient care.

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